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The misuse of the preparation system has left our team disillusioned. And for that reason, we decided it was time to make a difference. We had to make In this way, CertyIQ was created to provide quality materials without stealing from everyday people who are trying to make a living.

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John

October 19, 2022



Thanks you so much for your help. I scored 972 in my exam today. More than 90% were from your PDFs!

Dana

September 04, 2022



Thanks a lot for this updated AZ-900 Q&A. I just passed my exam and got 974, I followed both of your Az-900 videos and the 6 PDF, the PDFs are very much valid, all answers are correct. Could you please create a similar video/PDF for DP900, your content/PDF's is really awesome. The team did a really good job. Thank You 😊.

Ahamed Shibly

2 months ago



Customer support is really fast and helpful, I just finished my exam and this video along with the 6 PDF helped me pass! Definitely recommend getting the PDFs. Thank you!

October 22, 2022



Passed my exam today with 891 marks. Out of 52 questions, 51 were from certyiq PDFs including Contoso case study. Thank You certyiq team!

Henry Rome

2 months ago



These questions are real and 100 % valid. Thank you so much for your efforts, also your 4 PDFs are awesome, I passed the DP900 exam on 1 Sept. With 968 marks. Thanks a lot, buddy!

Esmaria

2 months ago



Simple easy to understand explanations. To anyone out there wanting to write AZ900, I highly recommend 6 PDF's. Thank you so much, appreciate all your hard work in having such great content. Passed my exam Today -3 September with 942 score.

Nvidia

(NCP-AAI)

Agentic AI

Total: **182 Questions**

Link: <https://certyiq.com/papers/nvidia/ncp-ai>

Question: 1**CertyIQ**

When designing tool integration for an agent that needs to perform mathematical calculations, web searches, and API calls, which architecture pattern provides the most scalable and maintainable approach?

- A. External tool services with manual configuration for each agent instance
- B. Microservice-based tool architecture with standardized interfaces
- C. Monolithic tool handler with conditional logic for different tool types
- D. Embedded tool functions within the main agent code

Answer: B**Question: 2****CertyIQ**

A company is deploying an AI-powered customer support agent that integrates external APIs and handles a wide range of customer inputs dynamically.

Which of the following strategies are appropriate when designing an AI agent for dynamic conversation management and external system interaction? (Choose two.)

- A. Integrating a feedback loop from user interactions to iteratively improve agent behavior.
- B. Using rule-based logic as the primary framework to maintain consistency in agent decisions.
- C. Implementing retry logic for API failures to ensure robustness in external communications.
- D. Preferring hardcoded responses for frequent queries to deliver reliable and low-latency answers.

Answer: AC**Question: 3****CertyIQ**

In the context of agent development, how does an autonomous agent differ from a predefined workflow when applied to complex enterprise tasks?

- A. Agents optimize for execution speed under fixed input-output mappings, while workflows prioritize goal alignment through adaptive reasoning and memory mechanisms.
- B. Workflows provide deterministic task sequencing with conditional branching, while agents adapt decisions dynamically based on goals, context, and environment feedback.
- C. Workflows emphasize parallelism and distributed coordination of processes, while agents emphasize serialization and isolated problem solving.

Answer: B**Question: 4****CertyIQ**

A Lead AI Architect at a global financial institution is designing a multi-agent fraud detection system using an agentic AI framework. The system must operate in real time, with distinct agents working collaboratively to monitor and analyze transactional patterns across accounts, retain and share contextual information over time, and escalate suspicious behaviors to a human fraud analyst when needed.

Which architectural approach enables intelligent specialization, shared memory, and inter-agent coordination in a dynamic and evolving threat environment?

- A. Design a modular multi-agent system where individual agents collaborate asynchronously using shared memory and structured messaging.
- B. Design a multi-agent system where individual agents collaborate synchronously using shared memory and structured messaging.
- C. Design a centralized rule-based service that checks all transactions against static fraud indicators and sends alerts when thresholds are exceeded.
- D. Design an agentic workflow where each agent acts independently on isolated data slices with no inter-agent communication to reduce latency and model complexity.
- E. Design monolithic LLM-based agents that handle all fraud detection tasks within a single loop, without modular roles or multi-agent coordination.

Answer: A

Question: 5

CertyIQ

When designing complex agentic workflows that include both sequential and parallel task execution, which orchestration pattern offers the greatest flexibility?

- A. Graph-based workflow orchestration incorporating conditional branches
- B. Linear pipeline orchestration with a fixed task sequence
- C. Event-driven orchestration that triggers tasks reactively, in series or in parallel

Answer: A

Question: 6

CertyIQ

When implementing inter-agent communication for a distributed agentic system running across multiple NVIDIA GPU nodes, which message routing pattern provides the best balance of reliability and performance?

- A. Database-based message queuing with polling
- B. Direct TCP connections between all agent pairs
- C. Event-driven message routing with distributed broker clusters
- D. Centralized message broker with topic-based routing

Answer: C

Question: 7

CertyIQ

Which two orchestration methods are MOST suitable for implementing complex agentic workflows that require both external data access and specialized task delegation? (Choose two.)

- A. Agentic orchestration with specialized expert system delegation

- B. Prompt chaining to accomplish state management
- C. Manual workflow coordination without automation
- D. Retrieval-based orchestration for external data
- E. Static rule-based routing with predefined pathways

Answer: AD

Question: 8

CertyIQ

When evaluating coordination failures in a multi-agent system managing distributed manufacturing workflows, which analysis approach best identifies state management and planning synchronization issues?

- A. Monitor agent outputs individually to confirm local correctness and examine results of specific workflow steps.
- B. Deploy distributed state tracing across agents, analyze transition timing, study communication overhead, and verify synchronization accuracy.
- C. Assess synchronization methods during design reviews and use simulations to evaluate coordination across representative workflow scenarios.
- D. Track workflow throughput and task completions to measure performance trends and highlight workflow outcomes.

Answer: B

Question: 9

CertyIQ

You are designing an AI agent for summarizing medical documents that include images and text as well. It must extract key information and recognize dates.

Which feature is most critical for ensuring the agent performs well across multiple input and output formats?

- A. Use of guardrails to filter out hallucinated content
- B. Retry logic implementation to ensure robustness during API failures
- C. Chain-of-thought prompting for reasoning accuracy
- D. Multi-modal model integration to handle both text and vision inputs

Answer: D

Question: 10

CertyIQ

Which two coordination patterns are MOST effective for implementing a multi-agent system where agents have different specializations (Research Analyst, Content Writer, Quality Validator)?

- A. Sequential pipeline coordination with crew-based structured handoffs
- B. Peer-to-peer coordination with consensus mechanisms
- C. Random task distribution with load balancing

D. Hierarchical coordination with crew-based task delegation

Answer: AD

Question: 11

CertyIQ

A senior AI architect at a public electricity utility is designing an AI system to automate grid operations such as outage detection, load balancing, and escalation handling. The system involves multiple intelligent agents that must operate concurrently, respond to changing data in real time, and collaborate on tasks that evolve over multiple interaction steps. The architect must choose a design pattern that supports coordination, flexible task delegation, and responsiveness without sacrificing maintainability.

Which design approach is most appropriate for this scenario?

- A. Use an agent service architecture with decoupled execution units managed by a shared interface layer that handles communication and task routing.
- B. Build a rule-driven control structure that maps task flows to predefined paths for fast and efficient execution under known operating conditions.
- C. Design the system as a stepwise sequence of agent functions, where each stage processes and passes data to the next in a fixed functional chain.
- D. Adopt a role-based agent model coordinated through a shared task planner, where agent decisions are informed by centralized policy logic and runtime context signals.

Answer: D

Question: 12

CertyIQ

An AI engineer is evaluating an underperforming multi-agent workflow built with NVIDIA agentic frameworks. Which analysis approach most effectively identifies optimization opportunities in agent coordination and communication patterns?

- A. Monitor workflow completion times using analysis that subsumes inter-agent communication costs, coordination overhead, and task allocation balance.
- B. Focus exclusively on individual agent accuracy without analyzing workflow-level efficiency, coordination costs, or overall system throughput.
- C. Evaluate agents individually, allowing the toolkit to automatically infer interaction effects, communication patterns, and emergent behaviors from coordination.
- D. Trace agent interaction patterns using observability features, measure communication overhead, identify redundant operations, and analyze task distribution efficiency.

Answer: D

Question: 13

CertyIQ

You are designing a virtual assistant that helps users check weather updates via external APIs. During testing, the agent frequently calls the incorrect tools, often hallucinating endpoints or returning incorrect formats. You suspect the prompt structure might be the root cause of these failures.

Which prompt design best supports consistent tool invocation in this agent?

- A. Rely on the agent's internal knowledge to infer tool usage
- B. Include tool names in natural language but without parameter examples
- C. Provide only a generic system instruction with no examples
- D. Use structured prompt templates with few-shot tool usage examples

Answer: D

Question: 14

CertyIQ

You're working with an LLM to automatically summarize research papers. The summaries often omit critical findings.

What's the best way to ensure that the summaries accurately reflect the core insights of the research papers?

- A. Asking the LLM to "summarize the paper."
- B. Asking the LLM to "understand" the paper to generate a summary.
- C. Having the LLM generate the summaries and then manually review every output.
- D. Asking the LLM to "extract the key findings."

Answer: D

Question: 15

CertyIQ

When implementing tool orchestration for an agent that needs to dynamically select from multiple tools (calculator, web search, API calls), which selection strategy provides the most reliable results?

- A. Random dynamic tool selection with retry mechanisms and usage examples
- B. LLM-based tool selection with structured tool descriptions and usage examples
- C. Rule-based selection with predefined tool mappings and usage examples
- D. Configuration-based tool selection with manual specifications and usage examples

Answer: B

Question: 16

CertyIQ

An engineer has created a working AI agent solution providing helpful services to users. However, during live testing, the AI agent does not perform tasks consistently.

Which two potential solutions might help with this issue? (Choose two.)

- A. Remove schema validations and assertions on tool outputs to avoid inconsistency.
- B. Increase randomness (e.g., temperature) and remove fixed seeds to avoid determinism.
- C. Identify where dividing the tasks into subtasks and handling them by multiple agents can help.
- D. Refine the prompt given to the AI Agent; be clear on objectives

Answer: CD

Question: 17

CertyIQ

A development team is building a customer support agent that interacts with users via chat. The agent must reliably fetch information from external databases, handle occasional API failures without crashing, and improve its responses by learning from user feedback over time.

Which of the following tasks is most critical when enhancing an AI agent to handle real-world interactions and improve over time?

- A. Applying a well-structured training process with foundational generative models and prompt engineering
- B. Utilizing internal knowledge bases to support agent responses alongside external APIs
- C. Implementing retry logic for error handling and integrating user feedback loops for iterative improvement
- D. Designing conversation flows that provide consistent responses based on predefined scripts

Answer: C

Question: 18

CertyIQ

What NVIDIA framework can be used to train a better agent?

- A. NeMo-RL
- B. NeMo Guardrails
- C. TensorRT-LLM

Answer: A

Question: 19

CertyIQ

You are evaluating your RAG pipeline. You notice that the LLM-as-a-Judge consistently assigns high similarity scores to responses that contain irrelevant information.

What should you investigate as the most likely potential cause with the least development effort?

- A. The temperature setting used by the LLM during response generation.
- B. The size of the knowledge base used to power the RAG pipeline.
- C. The quality of the synthetic questions used for evaluation.
- D. The prompt used to instruct the LLM-as-a-Judge to assess the response.

Answer: D

Question: 20

CertyIQ

You're managing an agentic AI responsible for customer support ticket triage. The agent has been consistently accurate in routing tickets to the appropriate departments. However, a team leader has noticed a significant

increase in the number of tickets requiring “escalation” – cases where the agent initially misclassified a complex issue as a simple, routine one, leading to delays and frustrated customers.

What would be an appropriate first step in resolving this issue?

- A. Analyzing the agent’s decision-making process, focusing on the specific criteria it uses to classify tickets, and identifying potential biases or blind spots.
- B. Adjusting the agent’s reward function to prioritize speed of resolution over accuracy, as a first step in analysis of the problem.
- C. Increasing the agent’s autonomy, granting it more decision-making power during triage to improve its efficiency.
- D. Conducting a “red-teaming” exercise, having human agents deliberately create complex and ambiguous scenarios to analyze the agent’s robustness.

Answer: A

Question: 21

CertyIQ

A customer service agentic AI is designed to resolve billing inquiries. It consistently resolves inquiries accurately and efficiently. However, a significant number of customers are reporting frustration due to the agent’s tendency to repeatedly ask for the same information (account number, address) during each interaction, even after it’s already been provided.

Which evaluation method would be most effective for addressing this issue?

- A. Adjusting the agent’s reward function to prioritize speed of resolution over customer satisfaction.
- B. Analyzing the agent’s dialogue transcripts to identify patterns in its questioning techniques.
- C. Implementing a “conversational flow” analysis to optimize the order of questions asked during each interaction.
- D. Increasing the agent’s processing speed to reduce the time it takes to handle each inquiry and increase customer satisfaction.

Answer: B

Question: 22

CertyIQ

A financial services agentic AI is being used to automate initial customer onboarding. The agent is completing the process efficiently and accurately, but reviews of its conversations reveal it often uses overly formal and complex language that confuses customers.

Which type of evaluation is best suited to address this issue?

- A. Controlled user testing sessions to collect user feedback on the clarity and tone of responses
- B. Compliance review of the agent’s access to regulatory guidelines and policy documentation
- C. Continuous user feedback collection, specifically gathering subjective assessments of the agent’s communication style
- D. Statistical analysis of the agent’s decision-making patterns to detect overly formal and complex response choices

Answer: A

Question: 23**CertyIQ**

You're evaluating the performance of a tool-using agent (e.g., one that issues API calls or executes functions). From the list below, what are two important features to evaluate? (Choose two.)

- A. Tool use accuracy
- B. Tokens per second
- C. Tool use rate
- D. Task completion rate

Answer: AD**Question: 24****CertyIQ**

When analyzing user feedback patterns to improve a technical documentation agent, which evaluation methods effectively translate feedback into actionable optimization strategies? (Choose two.)

- A. Collect broad user feedback as-is, enabling rapid accumulation of suggestions and diverse perspectives for potential future analysis.
- B. Design iterative feedback loops with version tracking, A/B testing of improvements, and regression monitoring to ensure changes enhance rather than degrade performance
- C. Incorporate user suggestions rapidly to maximize responsiveness and demonstrate continuous adaptation to evolving user needs.
- D. Implement feedback categorization systems grouping issues by type (accuracy, clarity, completeness) with quantitative impact scoring and improvement prioritization matrices

Answer: BD**Question: 25****CertyIQ**

When analyzing an agent's failure to complete multi-step financial analysis tasks, which evaluation approach best identifies prompt engineering improvements needed for reliable task decomposition and execution?

- A. Implement systematic prompt testing with chain-of-thought reasoning templates, step-by-step decomposition analysis, and success rate tracking across tasks of varying complexity.
- B. Focus primarily on response speed optimization as a primary focus over reasoning quality, step completion accuracy, and prompt clarity for complex analytical requirements.
- C. Test only final output accuracy as this will automatically include intermediate reasoning steps, decomposition quality, and prompt structure effectiveness for complex workflows.
- D. Rely on generic prompt templates which are by default already optimized for general use, instead of tailoring them to financial terminology, calculation needs, or specialized multi-step analysis patterns.

Answer: A

Question: 26**CertyIQ**

An agentic AI is tasked with generating marketing copy for various campaigns. It's consistently producing high-quality text and generating significant engagement. However, qualitative feedback from brand managers indicates that the content lacks a distinct "brand voice" and feels generic.

Which of the following metrics would be most valuable for evaluating the agent's adherence to the brand's established voice?

- A. A metric assessing the agent's ability to tailor its language and messaging for distinct audience segments based on demographic and psychographic data.
- B. A metric evaluating the agent's textual similarity to a formalized brand style guide, analyzing factors such as tone, approved vocabulary, and prescribed sentence structures.
- C. A metric tracking the average word count and sentence length of the agent's copy, focusing on stylistic efficiency as a potential proxy for brand alignment.
- D. A metric quantifying how frequently the agent's output is shared, liked, or reposted on major social platforms, using this as an indicator of effective brand representation.

Answer: B**Question: 27****CertyIQ**

When analyzing suboptimal agent response quality after deployment, which parameter tuning evaluation methods effectively identify the optimal configuration adjustments? (Choose two.)

- A. Design ablation studies systematically varying individual parameters while holding others constant to isolate each parameter's impact on agent behavior and performance.
- B. Apply identical parameter settings across all agent types and tasks, promoting consistency and simplifying comparison across different use cases.
- C. Implement A/B testing frameworks comparing temperature, top-k, and top-p variations while measuring task-specific quality metrics and user satisfaction scores.
- D. Use production traffic directly for parameter experiments, enabling real-world insights and faster identification of impactful settings.
- E. Randomly adjust all parameters simultaneously, allowing for broader exploration of the parameter space in a shorter time frame.

Answer: AC**Question: 28****CertyIQ**

You are tasked with comparing two agentic AI systems – System A and System B – both designed to generate marketing copy. You've run identical prompts and have recorded the generated outputs.

To objectively assess which system is performing better, what is the most appropriate approach?

- A. Measure the click-through rate for each system's marketing copy as the primary indicator of performance.
- B. Implement a human-in-the-loop to subjectively rate each output on a scale of 1 to 5 based on the user's personal preference.
- C. Implement a benchmark pipeline that automatically compares the generated outputs using metrics like relevance, creativity, and grammatical correctness.

D. Gather ratings from a panel of users, with each rating marketing copy on a 1 to 5 scale for overall impression of relevance, creativity, and grammatical correctness.

Answer: C

Question: 29

CertyIQ

You're evaluating the RAG pipeline by comparing its responses to synthetic questions. You've collected a large set of similarity scores.

What's the primary benefit of aggregating these scores into a single metric (e.g., average similarity)?

- A. Aggregation identifies the specific chunks within the RAG pipeline that are contributing to the highest similarity scores.
- B. Aggregation reduces the complexity of the evaluation process and allows for a more overall assessment of the pipeline's effectiveness.
- C. Aggregation provides a more accurate representation of the RAG pipeline's performance.
- D. Aggregation eliminates the need for qualitative analysis of the RAG pipeline's responses.

Answer: B

Question: 30

CertyIQ

In designing an AI workflow which of the following best describes a comprehensive approach to improving the performance of AI agents?

- A. Implementing benchmarking pipelines, deploying physical agents and monitoring user engagement metrics
- B. Implementing benchmarking pipelines, collecting user feedback, and tuning model parameters iteratively
- C. Implementing benchmarking pipelines and incorporating a dynamic dataset for a real-time fall-back
- D. Monitoring agents' throughput and time-to-first-token from the scoring engine

Answer: B

Question: 31

CertyIQ

You're employing an LLM to automate the generation of email responses for a customer service team. The generated responses frequently miss the mark, failing to address the customer's underlying concerns. What's the most crucial element to add to the prompt to enhance the quality of the email responses?

- A. Instructing the LLM with a detailed prompt containing instructions on how to format and compose the response in an easy-to-understand structure.
- B. Instructing the LLM to use a simple template for all email replies before generating a response.
- C. Instructing the LLM to "understand the customer's issue" before generating a response.
- D. Instructing the LLM to provide a response that "is the most helpful" before generating a response.

Answer: A

Question: 32

CertyIQ

After a series of adjustments in a supply chain agentic system, the agent has dramatically reduced shipping times and minimized costs, but the team is receiving a high volume of complaints from customers regarding delayed deliveries.

Which metric is MOST important to prioritize when investigating this situation?

- A. The agent's ability to predict future demand fluctuations, as accurate forecasting is crucial for effective logistics.
- B. The total cost savings achieved through the agent's optimization, which represents a significant financial benefit.
- C. The percentage of delivery times that fall within the acceptable delay window, considering customer satisfaction as a key factor.
- D. The agent's adherence to the prescribed delivery schedules, as it's demonstrably improving efficiency.

Answer: C

Question: 33

CertyIQ

A recently deployed Agentic AI system designed for automated incident response within a cloud infrastructure has been consistently failing to identify and resolve 'high-priority' alerts – specifically, those related to increased CPU utilization across several virtual machines. Initial logs show the agent is primarily focusing on alerts with related network traffic spikes, ignoring the CPU metrics.

What is the most appropriate initial step for a senior Agentic AI engineer to take to resolve this issue, considering the system's reliance on benchmarking and iterative improvement?

- A. Review the agent's evaluation framework, focusing on the defined benchmarks used to assess its response efficiency and impact on overall system performance.
- B. Replace the agent's underlying AI model with a more powerful, general-purpose machine learning engine as a first step in investigating current benchmarks.
- C. Implement a new synthetic data set containing a wide variety of CPU load profiles to train the agent's decision-making model.
- D. Review the agent's sensitivity thresholds, focusing on CPU utilization alerts to maximize detection accuracy.

Answer: A

Question: 34

CertyIQ

A team is evaluating multiple versions of an AI agent designed for customer support. They want to identify which version completes tasks more efficiently, responds accurately, and improves over time using user feedback.

Which practice is most important to ensure continuous refinement and optimal performance of the AI agent?

- A. Comparing agents on isolated tasks without standardized benchmarking pipelines
- B. Relying solely on offline benchmarks without incorporating live user feedback during tuning
- C. Implementing an evaluation framework that quantifies task efficiency and incorporates human-in-the-loop feedback

D. Tuning model parameters once before deployment to maximize initial accuracy

Answer: C

Question: 35

CertyIQ

When analyzing inconsistent performance across a fleet of customer service agents handling similar queries, which evaluation approach most effectively identifies root causes and optimization opportunities?

- A. Assess performance data from recently improved agents and highlight strong results, using outcome comparisons to identify areas with the greatest impact on service quality.
- B. Average performance metrics across all agents as this will smooth individual variations, query distribution differences, and temporal factors affecting agent behavior and accuracy.
- C. Deploy stratified evaluation sampling across agent variants, query complexity levels, and temporal patterns while tracking decision paths using comparative analytics.
- D. Review performance across both high- and low-accuracy agent groups, comparing case outcomes and identifying patterns contributing to top and bottom results.

Answer: C

Question: 36

CertyIQ

You are using an LLM-as-a-Judge to evaluate a RAG pipeline.

What is the primary benefit of synthetically generating question-answer pairs, rather than relying solely on human-created test cases?

- A. Synthetically generated questions are more challenging and reveal deeper flaws in the RAG pipeline.
- B. Synthetic generation eliminates the need for any human validation of the RAG pipeline's output.
- C. Synthetically generated answers are inherently more accurate than those produced by the LLM.
- D. Synthetic generation allows for systematic testing of the RAG pipeline across a wider range of scenarios and query types.

Answer: D

Question: 37

CertyIQ

Your agent is generating inconsistent and contradictory statements.

Which approach would be most suitable to improve the agent's output?

- A. Employing Reflexion
- B. Increasing the number of generated plans
- C. Using Decomposition-First Planning
- D. Decreasing the length of prompts

Answer: A

Question: 38

CertyIQ

You're utilizing an LLM to translate complex technical documentation into multiple languages. The translations often lack nuance and fail to capture the original intent.

What's the most effective strategy for improving the quality of the translations?

- A. Providing the LLM with a glossary of key terms, concepts in all languages and the dataset of previously translated text.
- B. Training the LLM on a dataset of translated texts.
- C. Providing the LLM with guidance to "translate the documents" without additional guidance, so it can use trained knowledge.
- D. Providing the LLM with guidance to translate "with high accuracy" without additional guidance, so it can use trained knowledge.

Answer: A

Question: 39

CertyIQ

An e-commerce platform is implementing an AI-powered customer support system that handles inquiries ranging from simple FAQ responses to complex product recommendations and technical troubleshooting. The system experiences unpredictable traffic patterns with sudden spikes during sales events and varying complexity requirements. Simple questions comprise the majority of requests but require minimal compute, while complex product recommendations need sophisticated reasoning. The company wants to optimize costs while maintaining service quality across all query types.

Which approach would provide the MOST cost-optimized scaling strategy for this variable-workload, mixed-complexity environment?

- A. Deploy specialized NVIDIA NIM microservices using a single large model configuration that handles all agent functions on high-capacity GPUs, with auto-scaling infrastructure that maintains constant resource allocation across all traffic patterns.
- B. Deploy specialized NVIDIA NIM microservices on CPU-optimized infrastructure with auto-scaling capabilities to minimize hardware costs, while accepting longer inference times for cost optimization benefits.
- C. Deploy specialized NVIDIA NIM microservices with an LLM router to dynamically route requests to appropriate models based on complexity, combined with auto-scaling infrastructure that scales different model types independently.
- D. Deploy multiple specialized NVIDIA NIM microservices with identical high-capacity models across all available GPUs, implementing auto-scaling infrastructure without request complexity differentiation or dynamic model selection capabilities.

Answer: C

Question: 40

CertyIQ

A technology startup is preparing to launch an AI agent platform to serve clients with unpredictable usage patterns. They face periods of high user activity and low demand, so their deployment approach must minimize wasted resources during slow times and automatically allocate more resources during busy periods – all while keeping operational costs reasonable.

Given these requirements, which deployment strategy most effectively ensures both cost-effectiveness and

adaptability for scaling agentic AI systems?

- A. Scheduling periodic manual reviews to increase or decrease infrastructure based on predicted user numbers
- B. Monitoring system logs for usage patterns and making infrastructure changes after monthly analysis
- C. Using fixed-size virtual machine clusters to guarantee consistent resource allocation at all times
- D. Implementing autoscaling policies in a container orchestration environment to automatically adjust resources according to workload changes

Answer: D

Question: 41

CertyIQ

When evaluating a multi-agent customer service system experiencing unpredictable scaling costs and performance bottlenecks during peak hours, which analysis approaches effectively identify optimization opportunities for both infrastructure efficiency and service reliability? (Choose two.)

- A. Maintain consistent resource allocation across all service hours, for a more precise view of baseline traffic impact on long-term infrastructure efficiency.
- B. Scale agent infrastructure based on aggregate performance trends, using system-wide monitoring tools to identify broader optimization patterns across resources.
- C. Deploy agents with configurable scaling workflows, allowing analysis of resource adjustment strategies and their effects on service stability during variable demand periods.
- D. Deploy distributed tracing with cost attribution per agent type, correlating resource consumption with business value metrics to identify optimization opportunities in agent deployment strategies.
- E. Implement comprehensive workload profiling using NVIDIA Nsight to analyze GPU utilization patterns, identify underutilized resources, and optimize batch sizing for dynamic scaling with Kubernetes HPA.

Answer: DE

Question: 42

CertyIQ

When analyzing throughput bottlenecks in a multi-modal agent processing text, images, and audio, which Triton configuration evaluations identify optimization opportunities? (Choose two.)

- A. Analyze model ensemble pipelines for sequential dependencies, identify parallelization opportunities, and optimize inter-model data transfer using Triton's scheduler.
- B. Profile GPU memory allocation patterns across modalities, implement model instance batching strategies, and tune concurrency limits to maximize utilization.
- C. Deploy each modality on separate Triton instances, allowing Triton to automatically manage ensemble coordination, shared memory usage, and pipeline integration.
- D. Use a single model instance per GPU, allowing Triton to automatically optimize concurrency, batching, and multi-instance settings for throughput scaling.

Answer: AB

Question: 43**CertyIQ**

When analyzing performance bottlenecks in a multi-modal agent processing customer support tickets with text, images, and voice inputs, which evaluation approach most effectively identifies optimization opportunities?

- A. Measure total response time as this analyzes aggregated performance trends across modalities, model loading times, and opportunities for parallel execution.
- B. Profile end-to-end latency across modalities, measure model switching overhead, analyze batch processing opportunities, and evaluate Triton's dynamic batching for multi-modal workloads.
- C. Optimize each modality independently using dedicated profiling of cross-modal interactions, shared resource constraints, and pipeline execution strategies.
- D. Extend evaluation to accuracy and quality metrics, incorporating resource usage patterns, latency observations, and their impact on user experience.

Answer: B**Question: 44****CertyIQ**

What benefits does a Kubernetes deployment offer over Slurm?

- A. Kubernetes provides autoscaling, auto-restarts, dynamic task scheduling, error isolation with containers, and integrated monitoring.
- B. Kubernetes is the best option for both training and inference, offering advantages for resource management and workload visibility over traditional HPC schedulers like Slurm.
- C. Kubernetes is more optimized for batch jobs to achieve high throughput, and also provides for monitoring and failover in large-scale workloads.

Answer: A**Question: 45****CertyIQ**

A company plans to launch a multi-agent system that must serve thousands of users simultaneously. The team needs to ensure the system remains reliable, scales efficiently as demand increases, and operates in a cost-effective manner.

Which approach is most effective for achieving robust and scalable deployment of an agentic AI system in production?

- A. Running agents without load balancing to reduce infrastructure complexity and achieve robust and scalable deployment of an agentic system
- B. Establishing a continuous monitoring framework to track system performance and adapt resources as usage patterns evolve
- C. Deploying all agents on a single server with ongoing performance monitoring to maximize hardware utilization
- D. Orchestrating agents using containerization platforms, combined with load balancing and ongoing performance monitoring

Answer: D

Question: 46**CertyIQ**

A social media company wants to expand its agentic system to support global users, minimize downtime, and ensure smooth operation during usage spikes. The team is considering various deployment and scaling strategies to achieve these goals.

Which solution most effectively supports reliable and scalable deployment for an agentic AI system serving a global user base?

- A. Integrating MLOps practices for continuous deployment and rapid model updates in production environments
- B. Designing a distributed system architecture with multi-region deployment, automated failover, and dynamic resource allocation
- C. Implementing containerization with Docker to simplify deployment and streamline updates
- D. Using hardware profiling to optimize agent workloads for efficient GPU utilization across all deployed instances

Answer: B**Question: 47****CertyIQ**

A company is deploying a multi-agent AI system to handle large-scale customer interactions. They want to ensure the system is highly available, cost-effective, and scalable across multiple NVIDIA GPUs using container orchestration tools.

Which practice is most crucial for successfully deploying and scaling an agentic AI system in production?

- A. Use a static assignment of requests across agents to maintain consistent agent operation and simplify coordination while scaling infrastructure resources as needed.
- B. Optimize GPU utilization frameworks with workload optimization separate from cost analysis, prioritizing resource performance for peak load scenarios in deployment.
- C. Deploy agents on a single machine to obtain a dimensioning baseline and thereby reduce setup complexity before expanding system scope.
- D. Implementing automated workload management and resource scheduling frameworks to optimize GPU utilization and maintain service availability.

Answer: D**Question: 48****CertyIQ**

You are deploying a multi-agent customer-support system on Kubernetes using NVIDIA GPU nodes and Triton Inference Server. Traffic spikes during product launches. You need <100ms response times, zero downtime, automatic GPU scaling, and full monitoring.

Which deployment setup best achieves cost-effective, reliable, low-latency scaling?

- A. Set up one mixed GPU node pool with Cluster Autoscaler min=0, scale by network throughput, monitor via metrics-server and logs, and skip readiness probes for fast startup.
- B. Place GPU pods on on-demand nodes in one zone, disable Cluster Autoscaler, run a fixed pod count for bursts, scale on CPU usage, and monitor with default health checks.
- C. Deploy GPU pods in a node pool spanning all zones, mix GPU types, enable Cluster and Horizontal Pod Autoscalers using Prometheus GPU and latency metrics, and monitor with NVIDIA DCGM and Grafana.

D. Use spot-instance node pools across zones, enable Cluster Autoscaler with capped nodes, scale on memory usage, and monitor with logs and cluster events.

Answer: C

Question: 49

CertyIQ

Which two deployment patterns are MOST suitable for scaling agentic workloads on NVIDIA Infrastructure? (Choose two.)

- A. Bare metal deployment with manual resource allocation
- B. Static virtual machine deployment with fixed resources
- C. Serverless deployment without GPU acceleration
- D. Containerized deployment with NIM (NVIDIA Inference Microservices)
- E. Kubernetes orchestration with Horizontal Pod Autoscaling (HPA)

Answer: DE

Question: 50

CertyIQ

When evaluating an agent's degrading response times under increasing load, which analysis approach most effectively identifies scalability bottlenecks and optimization opportunities?

- A. Track average response time while examining stage-by-stage processing metrics, resource usage trends, and potential components impacting scalability.
- B. Test at fixed, low load levels while using controlled stress scenarios to compare with performance under production-like traffic patterns.
- C. Profile each major system stage using distributed tracing, analyze GPU utilization with NVIDIA performance tools, and map queuing delays against varying workload patterns.
- D. Focus on model inference duration while also measuring preprocessing time, tool-calling latency, and response formatting in the end-to-end pipeline.

Answer: C

Question: 51

CertyIQ

A company operates agent-based workloads in multiple data centers. They want to minimize latency for users in different regions, maintain continuous service during infrastructure upgrades, and keep operational costs predictable.

Which deployment practice best supports low-latency, resilient, and cost-efficient agent operations at scale?

- A. Schedule regular agent downtime for system updates and operational recalibration.
- B. Implement geo-distributed deployments with rolling updates and resource usage monitoring.
- C. Prioritize high-performance GPUs for all agents in geo-distributed deployments.
- D. Apply static infrastructure allocation with centralized resource usage monitoring at a single data center.

Answer: B

Question: 52

CertyIQ

When implementing stateful orchestration for agentic workflows using LangGraph, which memory management approach provides the best balance of performance and context retention?

- A. Store complete conversation history in memory with periodic database syncing
- B. Implement rolling window memory with fixed conversation length limits
- C. Use session-ID based checkpointer with user-defined schema for selective state persistence

Answer: C

Question: 53

CertyIQ

An AI Engineer at an automotive company is developing an inventory restocking assistant for parts that must plan reordering of parts over multiple days, factoring in stock levels, predicted demand, and supplier lead time. Which approach best equips the agent for sequential decision-making?

- A. Reinforcement learning sequence model using only a custom PyTorch Decision Transformer
- B. Rule-based reorder strategy with fixed thresholds implemented via NVIDIA Triton Inference Server
- C. Hybrid supervised/RL-trained model using NeMo-Aligner for policy alignment
- D. Reinforcement learning sequence model such as NVIDIA'S NeMo-RL framework

Answer: D

Question: 54

CertyIQ

An AI Engineer at a retail company is developing a customer support AI agent that needs to handle multi-turn conversations while keeping track of customers' previous queries, preferences, and unresolved issues across multiple sessions.

Which approach is most effective for managing context retention and enabling the agent to respond coherently in real time?

- A. Use a sliding window of recent conversation tokens in memory to track only the last few exchanges.
- B. Retrain the model periodically using historical logs to improve long-term contextual understanding.
- C. Implement a hybrid memory system with vector-based search and key-value storage to retrieve relevant past interactions.
- D. Increase the maximum context window size so the full conversation history is processed each time.

Answer: C

Question: 55**CertyIQ**

An AI engineer at an oil and gas company is designing a multi-agent AI system to support drilling operations. Different agents are responsible for subsurface modeling, risk analysis, and resource allocation. These agents must share operational context, reason through interdependent planning steps, and justify their collaborative decisions using structured, transparent logic. The architecture must support memory persistence, sequential decision-making and chain-of-thought prompting across agents. Which implementation best supports this design?

- A. Orchestrate NeMo agents via Triton, use vector memory for shared context, ReAct planning, and NeMo Guardrails for reasoning.
- B. Use stateless LLM endpoints behind an API gateway and pass shared prompts across agents to simulate context and reasoning.
- C. Use LangChain to coordinate third-party agent APIs and store shared information in external memory, with logic encoded in static prompt chains.
- D. Fine-tune separate NeMo models for each agent role using LoRA, with pre-scripted action flows deployed via TensorRT for latency reduction.

Answer: A**Question: 56****CertyIQ**

In a global financial firm, an AI Architect is building a multi-agent compliance assistant using an agentic AI framework. The system must manage short-term memory for multi-turn interactions and long-term memory for persistent user and policy context. It should enable contextual recall and adaptation across sessions using NVIDIA's tool stack. Which architectural approach best supports these requirements?

- A. Leverage NVIDIA NeMo Framework with modular memory management, integrating conversational state tracking, knowledge graphs, and vector store retrieval, while using LoRA-tuned models to adapt responses overtime.
- B. Leverage RAPIDS cuDF for memory tracking by streaming multi-turn conversation logs as GPU-resident data frames, assuming transactional history can be recalled and reasoned over using dataframe operations.
- C. Rely exclusively on TensorRT to encode all prior knowledge into compiled model weights, allowing inference-only execution with no external memory dependencies across sessions.
- D. Leverage NVIDIA Triton Inference Server with dynamic batching to cache session-level inputs between inference calls, and use an external Redis store for long-term memory.

Answer: A**Question: 57****CertyIQ**

You are creating a virtual assistant agent that needs to handle an increasingly wide range of tasks over an extended period. What is the primary benefit of combining external storage (like RAG) with fine-tuning (embodied memory) in this context?

- A. To enhance long-term reasoning capabilities and adaptability
- B. To accelerate the agent's initial response time
- C. To ensure the agent doesn't make any errors

D. To eliminate the need for external knowledge

Answer: A

Question: 58

CertyIQ

A development team is building an AI agent capable of autonomously planning and executing multi-step tasks while retaining context and learning from past interactions. Which practice is most important to enable the agent to effectively manage long-term memory and complex tasks?

- A. Implement memory mechanisms for context retention and apply chain-of-thought prompts to enhance reasoning.
- B. Use basic rule-based decision methods that emphasize fast responses over adaptive planning.
- C. Apply short-term memory approaches that handle each interaction independently of previous ones.
- D. Reduce planning features and memory management to keep the system streamlined.

Answer: A

Question: 59

CertyIQ

You are developing an agent that needs to perform a complex set of tasks repeatedly. Why is periodic fine-tuning an important aspect of long-term knowledge retention for this type of agent?

- A. It prevents the agent from becoming overly specialized to a single task.
- B. It eliminates the need for external storage like RAG.
- C. It prevents the agent from forgetting past successes and failures.
- D. It guarantees the agent will produce the same output for the same input.

Answer: C

Question: 60

CertyIQ

An agent is tasked with solving a series of complex mathematical problems that require external tools to find information. It often struggles to keep track of intermediate steps and reasoning. Which prompting technique would be MOST effective in improving the agent's clarity and reducing errors in its reasoning?

- A. ReAct
- B. Symbolic Planning
- C. Zero-shot CoT
- D. Multi-Plan Generation

Answer: A

Question: 61**CertyIQ**

When analyzing memory-related performance degradation in agents handling extended customer support sessions, which evaluation methods effectively identify optimization opportunities for context retention? (Choose two.)

- A. Clear memory after each interaction and reset session state, removing historical context needed for personalized tasks to identify optimization opportunities.
- B. Profile memory access patterns by measuring retrieval latency, relevance scoring accuracy, and storage efficiency while monitoring context window utilization to identify optimization opportunities.
- C. Use fixed memory allocation including all conversation types, topic changes, and user needs, allowing adaptive-free observation of interaction patterns to identify optimization opportunities.
- D. Implement sliding window analysis comparing context compression strategies, summarization quality, and information preservation rates across varying conversation lengths to identify optimization opportunities.
- E. Store all conversation history including all interactions, allowing adaptive-free observation of data to identify optimization opportunities.

Answer: BD**Question: 62****CertyIQ**

A team is designing an AI assistant that helps users with travel planning. The assistant should remember user preferences, build personalized itineraries, and update plans when users provide new requirements. Which approach best equips the AI assistant to provide personalized and adaptive travel recommendations?

- A. Using a single-step question-answering system enhanced with session-level keyword tracking to improve relevance during ongoing interactions.
- B. Designing the assistant to handle each user request independently, while using implicit signals within each session to suggest relevant options.
- C. Engineering multi-step reasoning frameworks with persistent memory systems to store and utilize user preferences.
- D. Providing the same set of travel options to every user but sorting them based on recent popular destinations.

Answer: C**Question: 63****CertyIQ**

Which memory architecture is most appropriate for an agent that must track conversation flow and remember user preferences across multiple interactions?

- A. Implement shared memory using NVSHMEM for short- and long-term context
- B. Single unified memory store with time-based expiration policies
- C. Hierarchical memory with separate short-term and long-term layers
- D. Distributed memory with full replication across all nodes

Answer: C

Question: 64

CertyIQ

Implement Memory Systems for Contextual Awareness

An enterprise AI system needs to maintain contextual information over multiple interactions with users. Which memory implementation approach would be MOST effective for managing both immediate context and long-term historical interactions within an agentic workflow?

- A. Rely predominantly on the context window of the base LLM model to store all historical interactions with minimal external memory supplementation.
- B. Implement a hybrid memory system with short-term memory for immediate context and a vector database for long-term memory with semantic retrieval capabilities.
- C. Use a static prompt template with fixed context for all interactions, thereby providing memory information in that form across conversation sessions.
- D. Store all user interactions in a simple key-value database which will by default provide organization and retrieval strategy for historical context management.

Answer: B

Question: 65

CertyIQ

An enterprise wants their AI agent to support complex project management tasks. The agent should remember ongoing project details, adjust its plans based on new information, and break down large goals into actionable steps.

Which strategy best enables the AI agent to autonomously decompose tasks and adapt to new information over time?

- A. Predefining static workflows for each project type to guarantee consistent execution
- B. Developing long-term knowledge retention strategies and dynamic state management for adaptive planning
- C. Storing recent user interactions in a temporary cache for immediate retrieval
- D. Applying rule-based logic to each new request isolated from previous project data

Answer: B

Question: 66

CertyIQ

In a ReAct (Reasoning-Acting) agent architecture, what is the correct sequence of operations when the agent encounters a complex multi-step problem requiring external tool usage?

- A. Thought --> Answer --> Action --> Observation
- B. Action --> Thought --> Observation --> Action --> Thought --> Observation --> Answer
- C. Observation --> Thought --> Action --> Observation --> Thought --> Action --> Answer
- D. Thought --> Action --> Observation --> Thought --> Action --> Observation --> Answer

Answer: D

Question: 67

CertyIQ

What is a key limitation of Chain-of-Thought (CoT) prompting when using smaller language models for reasoning tasks?

- A. CoT prompting simplifies error analysis for small models, making it easy to identify and correct mistakes at each reasoning step.
- B. CoT prompting ensures step-by-step outputs, enabling even small models to solve complex problems reliably.
- C. CoT prompting requires relatively large models; smaller models may produce reasoning chains that appear logical but are actually incorrect, leading to poorer performance.
- D. CoT prompting consistently improves the logical accuracy of outputs for both small and large language models.

Answer: C

Question: 68

CertyIQ

In a production agentic system handling thousands of concurrent conversations, which state management strategy provides optimal performance while ensuring context preservation?

- A. Global shared state with locks for concurrent access
- B. Session-isolated state with serialization and lazy loading
- C. Stateless design with context reconstruction from message history

Answer: B

Question: 69

CertyIQ

A company is building an AI agent that must retrieve information from large document collections and client databases in real time. The team wants to ensure fast, accurate retrieval and maintain high data quality. Which approach best supports efficient knowledge integration and effective data handling for such an agent?

- A. Using traditional relational databases because they don't need specialized retrieval mechanisms for all data queries
- B. Integrating client data sources as they already incorporate data quality checks or augmentation to speed up deployment
- C. Relying on pre-trained models instead of connecting to external knowledge sources during inference
- D. Implementing retrieval-augmented generation (RAG) pipelines combined with vector databases to accelerate access to relevant information

Answer: D

Question: 70**CertyIQ**

What is RAG Fusion primarily designed to achieve?

- A. Creating a separate, dedicated database for storing all the retrieved chunks.
- B. Minimizing the need for retrieval, allowing the LLM to generate responses directly from its internal knowledge.
- C. Blending information from multiple retrieved chunks into a single response generated by the LLM.
- D. Automatically translating and integrating all retrieved chunks into a single language.

Answer: C**Question: 71****CertyIQ**

Your agent's primary task is to collect data from a device management platform API. This API frequently returns data in JSON format, but the underlying schemas can evolve without detailed documentation. What's the most effective strategy for the agent to maintain consistent access to the data, despite potential schema changes?

- A. Building a JSON schema validator that constantly monitors and enforces the current API schema.
- B. Manually inspecting the API response's structure and updating the agent's code when a change is detected.
- C. Building a fixed data extraction method based on the API's historical response patterns.
- D. Building a flexible data mapping layer that can adapt to changes in the API response structure.

Answer: D**Question: 72****CertyIQ**

You are improving your Retrieval-Augmented Generation (RAG) pipeline, and you want to exploit the properties of the embedding model to improve the solutions. Within a RAG system, why is reranking retrieved chunks a crucial step after initial retrieval?

- A. Reranking improves retrieval accuracy, regardless of the initial retrieval method.
- B. Reranking's purpose is to reduce the overall length of the response generated by the LLM.
- C. Reranking allows for prioritizing chunks based on relevance to the semantic intent of the query.
- D. Reranking primarily focuses on optimizing the visual presentation of the retrieved chunks.

Answer: C**Question: 73****CertyIQ**

In your RAG deployment, you've identified a performance bottleneck in the retrieval phase – specifically, the time it takes to access the vector database. Which of the following optimization strategies is most aligned with micro-service best practices, considering your RAG architecture?

- A. Implement a “cache-and-check” mechanism where the retrieval microservice immediately returns the first matching chunk, regardless of relevance.
- B. Increase the size of the LLM model itself, because it will automatically accelerate the overall response time.
- C. Introduce a dedicated service responsible solely for querying the vector database and returning relevant chunks.
- D. Optimize the LLM prompt to be shorter and more concise, significantly reducing the computational load.

Answer: C

Question: 74

CertyIQ

Your agent is designed to manage tasks through a service management API. The API responds with detailed event logs, but these logs contain both metadata and structured data.

To ensure the agent correctly interprets and processes the data from these logs, what’s the most prudent approach?

- A. Employ a specialized parser that adheres to the API’s documentation, to insure strict adherence to structured data.
- B. Employing a modular design that allows the agent to dynamically adjust its parsing logic.
- C. Using a human-in-the-loop approach, manually inspecting and interpreting each log entry.
- D. Employ a specialized parser that extracts all data fields, regardless of their type.

Answer: B

Question: 75

CertyIQ

Your deployed legal assistant shows great performance but occasionally repeats incorrect legal terms. Which tuning method best improves factual reliability?

- A. Replace retrieval with static hard-coded text snippets
- B. Use more verbose prompts to reinforce correct definitions
- C. Increase output randomness to improve exploration
- D. Add fact-checking steps using external tools during generation

Answer: D

Question: 76

CertyIQ

An AI Engineer is experimenting with data retrieval performance within a RAG system.

Which of the following techniques is most likely to improve the quality of the retrieved chunks?

- A. Adding clarifying keywords and synonyms to the original query to broaden the search.
- B. Truncating long queries to fit within the LLM’s context window.

- C. Using a single, highly specific keyword to guarantee a precise match.
- D. Directly feeding the original query to the LLM without any modification.

Answer: A

Question: 77

CertyIQ

You're building a RAG system that uses RAG Fusion.

Which of the following approaches would be most effective in determining how to combine information from multiple retrieved chunks?

- A. Filtering out chunks considered inconsistent with others before presenting information to the LLM.
- B. Using the LLM to automatically identify the most important sentences within each chunk and combine them.
- C. Manually selecting the most relevant sentences from each chunk and inserting them into the LLM prompt.
- D. Concatenating the text from all retrieved chunks into a single block to form the response.

Answer: B

Question: 78

CertyIQ

You are designing the architecture for a RAG (Retrieval-Augmented Generation) system, and you are concerned about ensuring data freshness and minimizing latency.

Which of the following is the most important consideration when designing the architecture?

- A. Employing a consolidated architecture with a large service handling all data retrieval and LLM interaction. This ensures consistent performance and simplifies debugging.
- B. Using a synchronous, block-level approach, where the LLM continuously monitors the database for updates and retrieves the entire dataset with each prompt.
- C. Implementing a single, centralized database for all data, updated with a synchronous polling mechanism for the LLM to retrieve the latest information.
- D. Use a loosely coupled, event-driven micro-service architecture where separate services handle data indexing, retrieval, and LLM prompting.

Answer: D

Question: 79

CertyIQ

You're developing an agent that monitors social media mentions of your brand. The social media platform's API returns data mentioning your brand with varying confidence scores that the brand was actually being mentioned, but these scores aren't consistently calibrated.

Considering the unreliability of these confidence scores, what's the most reliable way for the agent to insure it is truly processing media mentions of the brand?

- A. Using an approach that filters mentions with basic keyword search and removes those with exceptionally low confidence scores, relying on the API data as a first-pass filter.
- B. Using an approach that treats all mentions as equally reliable, regardless of their confidence scores, and

applies a uniform data processing workflow to minimize inconsistency.

C. Using a threshold-based approach, accepting mentions only if their confidence score exceeds a predefined level that aligns with typical thresholds used for well-calibrated APIs.

D. Using an approach that combines the agent's text analysis with the API's confidence score, weighing the agent's assessment more heavily when identifying mentions.

Answer: D

Question: 80

CertyIQ

Optimize agentic workflow performance with the NVIDIA Agent Intelligence Toolkit.

Your organization is building a complex multi-agent system that needs to connect agents built on different frameworks while maintaining optimal performance.

Which key features of the NVIDIA Agent Intelligence Toolkit would be MOST beneficial for this implementation?

A. The toolkit is limited to simple agent-to-agent communication but cannot orchestrate complex multi-agent workflows.

B. The toolkit provides framework-agnostic integration ensuring reusability of components.

C. The toolkit is designed exclusively for NVIDIA framework agents and cannot integrate with other frameworks.

D. The toolkit focuses primarily on agent development but lacks evaluation capabilities.

Answer: B

Question: 81

CertyIQ

Which two optimization strategies are MOST effective for improving agent performance on NVIDIA GPU infrastructure? (Choose two.)

A. Using multi-GPU coordination to distribute workloads, enabling higher throughput and efficiency for scaling agent tasks.

B. Applying TensorRT-LLM optimizations to reduce inference latency by improving kernel efficiency and memory usage.

C. Expanding GPU memory capacity to support larger models, assuming this alone guarantees meaningful performance improvements.

D. Manually tuning kernel launch parameters to optimize individual operations while overlooking overall pipeline performance dynamics.

Answer: AB

Question: 82

CertyIQ

You are rolling out a multimodal conversational agent on NVIDIA's stack: the model is containerized as a TensorRT-LLM engine, served via Triton Inference Server behind NIM microservices for routing and scaling, and protected by NeMo Guardrails for safety and compliance. During early testing, end-to-end latency exceeds your target budget, and you need to tune batching, model precision, and guardrail checks while maintaining both throughput and enforcement of safety policies.

Which configuration change is most effective for reducing latency under these constraints while still enforcing NeMo Guardrails policies?

- A. Quantize the TensorRT-LLM engine to FP16, tune Triton's dynamic batching, and integrate NeMo Guardrails alongside inference to run policy checks in parallel.
- B. Quantize the TensorRT-LLM engine to INT8, disable dynamic batching, and invoke Guardrails checks synchronously within the inference path.
- C. Deploy separate Triton servers for model inference and guardrail validation, routing requests sequentially and merging outputs at the application layer.
- D. Keep FP32 precision, increase batch size aggressively, and perform Guardrails checks in a downstream microservice after inference.

Answer: A

Question: 83

CertyIQ

A healthcare AI company is deploying diagnostic agents that process medical imaging and patient data. The system must deliver consistent sub-100ms inference times for critical diagnoses while supporting deployment across multiple hospital sites with different NVIDIA GPU configurations (from RTX 6000 workstations to DGX systems). The agents need to maintain high accuracy while being portable across different hardware environments and capable of running efficiently on various GPU memory configurations.

Which optimization strategy would deliver the BEST performance improvements while maintaining deployment flexibility across diverse NVIDIA hardware configurations?

- A. Deploy agents with NVIDIA CUDA-optimized Docker containers using a sequential inference architecture that processes each layer individually with GPU-to-CPU memory transfers between operations to avoid memory issues.
- B. Deploy agents using NVIDIA NIM containers with CPU-optimized inference to avoid GPU memory constraints and ensure consistent performance across different hospital infrastructure configurations.
- C. Deploy models using NVIDIA TensorRT optimization in their original FP32 precision format without any quantization or memory optimization, requiring 32GB+ GPU memory across all deployment sites.
- D. Deploy agents using model optimizations with post-training quantization with Nvidia NIM deployment for portable performance across different GPU platforms and memory configurations.

Answer: D

Question: 84

CertyIQ

You are tasked with deploying a multi-modal agentic system that must respond to user queries with minimal latency while maintaining guardrails for safe and context-aware interactions.

Which of the following configurations best leverages NVIDIA's AI stack to meet these requirements?

- A. Integrate NeMo Guardrails, configure NIM microservices for optimized inference, use TensorRT-LLM for deployment, and profile the system using Triton Inference Server with multi-modal support.
- B. Integrate NeMo Guardrails, use Omniverse to generate synthetic data, configure NIM microservices for optimized inference, use TensorRT-LLM for deployment, and profile the system using NeMo Agent Toolkit for multi-modal support.
- C. Use NeMo Guardrails for safety, deploy the model with Triton Inference Server using default settings, and rely on hardware accelerators like GPU/TPU inference for cost efficiency.
- D. Use NIM microservices for deployment, optionally use NeMo Guardrails unless one wants to minimize the

inference overhead.

Answer: A

Question: 85

CertyIQ

Integrate NeMo Guardrails, configure NIM microservices for optimized inference, use TensorRT-LLM for deployment, and profile the system using Triton Inference Server with multi-modal support. Which of the following strategies aligns with best practices for operationalizing and scaling such Agentic systems?

- A. Use Docker containers orchestrated by Kubernetes, implement MLOps pipelines for CI/CD, monitor agent health with Prometheus/Grafana.
- B. Deploy agents on bare-metal servers to maximize performance and avoid container overhead, using manual scripts for orchestration and monitoring.
- C. Deploy all agents on a single high-performance GPU node to reduce latency, and use cron jobs for periodic health checks and updates.
- D. Run agents as independent serverless functions to minimize infrastructure management, relying primarily on cloud provider auto-scaling and logging tools.

Answer: A

Question: 86

CertyIQ

When evaluating optimization opportunities between NeMo Guardrails, NIM microservices, and TensorRT-LLM in a production healthcare agent, which analysis approach best identifies optimization opportunities across the NVIDIA stack?

- A. Conduct stress testing of individual microservices and guardrails to measure peak throughput and determine theoretical performance limits of each module.
- B. Use default configurations to establish a deployment baseline, focusing on stability before conducting deeper performance profiling.
- C. Create end-to-end latency waterfalls that capture guardrail overhead, NIM queuing delays, and TensorRT optimization benefits while assessing overall pipeline efficiency.
- D. Tune each component individually, focusing primarily on local performance metrics with secondary attention to integration patterns.

Answer: C

Question: 87

CertyIQ

When evaluating GPU utilization inefficiencies in deploying Llama Nemotron models across A100 and H100 clusters, which approaches help identify optimal resource allocation strategies? (Choose two.)

- A. Allow Nemotron variants to profile actual workload characteristics and allocate resources based on observed demands.
- B. Profile resource utilization for each Nemotron variant and match models to appropriate GPU tiers.

- C. Allocate all agents to H100 GPUs, allowing resource profiles to automatically adjust for model size and computational requirements.
- D. Assess concurrent execution capabilities by employing multi-instance GPU partitioning for varying workload types.

Answer: BD

Question: 88

CertyIQ

A financial services company is deploying a multi-agent customer service system consisting of three specialized agents: a reasoning LLM for complex queries, an embedding agent for document retrieval, and a re-ranking agent for result optimization. The system experiences significant traffic variations, with peak loads during business hours (10x normal traffic) and minimal usage overnight. The company needs a deployment solution that can handle these fluctuations cost-effectively while maintaining sub-second response times during peak periods. Which NVIDIA infrastructure approach would provide the MOST cost-effective and scalable deployment solution for this variable-load multi-agent system?

- A. Deploy agents directly on individual NVIDIA RTX workstations without containerization or orchestration, relying on load balancers with round-robin for traffic distribution.
- B. Deploy each agent on dedicated NVIDIA DGX systems with manual scaling based on previous days traffic predictions and static resource allocation for peak loads.
- C. Deploy NVIDIA NIM microservices on Kubernetes with auto-scaling capabilities, utilizing NVIDIA NIM Operator for lifecycle management and horizontal pod autoscaling based on custom metrics.
- D. Deploy all agents on a single large GPU instance without containerization, scaling compute by upgrading to larger GPU instances when needed.

Answer: C

Question: 89

CertyIQ

Your team notices a spike in failed tool calls from a deployed workflow agent after a recent API schema update. The agent still returns outputs, but many are irrelevant or incomplete. Which maintenance task should be prioritized to restore accurate behavior?

- A. Reset the agent's long-term memory and reinitialize logs.
- B. Update the tool function specifications and re-test action sequences.
- C. Increase model temperature to encourage tool exploration.
- D. Reduce tool retrieval vector similarity threshold to broaden context.

Answer: B

Question: 90

CertyIQ

When evaluating an agent's integration with external tools and APIs for data retrieval and action execution, which analysis approaches effectively identify reliability and performance issues? (Choose two.)

- A. Implement comprehensive API call tracing with latency measurement, success rates per endpoint, and correlation analysis between tool failures and task completion.
- B. Use static API endpoints and parameters configured during development, allowing consistent and effective agent integration across predictable workflows.
- C. Connect to external APIs with standard procedures and monitor request and response exchanges to isolate the analysis of integration reliability and effectiveness.
- D. Design integration tests simulating API version changes, schema modifications, and backward compatibility scenarios to ensure reliable tool connections across updates.

Answer: AD

Question: 91

CertyIQ

A recently deployed agent sometimes outputs empty responses under heavy system load. Which system-level signal is most useful for diagnosing this issue?

- A. Number of tool function arguments returned per query
- B. Retrieval similarity thresholds in vector search
- C. GPU memory utilization and server-side inference logs
- D. Prompt injection detection rate over time

Answer: C

Question: 92

CertyIQ

An AI architect at a national healthcare provider is maintaining an agentic AI system. The system must monitor model and system performance in real time, raise alerts on failures or anomalies, manage version control and rollback of diagnostic models, and provide transparent insight into agent behavior during patient care workflows. Which operational approach best supports these requirements using the NVIDIA AI stack?

- A. Containerize each agent in NIM with basic health checks running on cron jobs, and manage version rollback by swapping prebuilt container images.
- B. Optimize all models with TensorRT and use periodic manual log reviews and NVIDIA shell scripts for detecting service anomalies and managing rollback.
- C. Deploy agent models on NVIDIA Triton Inference Server with Prometheus and Grafana for performance alerting, and manage model lifecycle via NGC and the Triton model repository.
- D. Expose agents as stateless NVIDIA API endpoints and monitor activity through application logs, with model versions tracked in a Git-based script repository.

Answer: C

Question: 93

CertyIQ

You are building a customer-support chatbot that fetches user account data from an external billing API. During testing, the API sometimes returns timeouts or 500 errors. You want the agent to be resilient-retrying when appropriate but failing gracefully if the service is down.

Which strategy best handles intermittent failures in API calls while still ensuring a good user experience?

- A. Retry requests with a consistent short delay after each failure and notify the user as each retry takes place.
- B. Implement exponential-backoff retries with a circuit breaker, and return a clear message to the user if all retries fail.
- C. Return a standard fallback message on failures to maintain conversation flow and reduce the risk of service interruptions for the user.
- D. Schedule retries using a fixed delay for all failure types, maintaining predictable timing and user notifications after each attempt.

Answer: B

Question: 94

CertyIQ

When evaluating a customer service agent's resilience to API failures and network issues, which analysis methods effectively identify weaknesses in error handling and retry mechanisms? (Choose two.)

- A. Analyze retry logic for exponential backoff patterns, retry limits, and circuit breaker integration to prevent cascading failures in distributed systems.
- B. Implement retry mechanisms that standardize recovery attempts across scenarios, emphasizing consistency in handling errors.
- C. Use fixed retry intervals to avoid the pitfalls of dynamic tuning, keeping retry timing consistent across different error conditions.
- D. Test under normal network conditions to establish baseline behavior, comparing results against production performance during degraded service scenarios.
- E. Conduct failure injection testing with varied error types (timeouts, rate limits, malformed responses) while monitoring recovery patterns and fallback behavior.

Answer: AE

Question: 95

CertyIQ

An autonomous vehicle company operates a multi-agent AI system across its fleet to process real-time sensor data, make driving decisions, and communicate with cloud infrastructure. The company needs fleet-wide monitoring to track GPU utilization, inference times, and memory usage, correlate performance with driving conditions and system load, and predict safety issues before they occur. Which monitoring and observability approach would BEST meet these fleet-scale, safety-critical requirements?

- A. Deploy NVIDIA NIM microservices with Prometheus integration, NVIDIA Nsight Systems profiling, and Kubernetes-native monitoring to provide detailed metrics, profiling, and container orchestration observability across the entire stack.
- B. Implement layered application monitoring with distributed tracing, synthetic transaction monitoring, and custom dashboards to capture complex dependencies, transaction flow, and service-level performance trends across the fleet.
- C. Implement comprehensive APM solutions with real-time baselines, automated root cause analysis, and fleet management integration to coordinate operational insights and performance management across thousands of vehicles.
- D. Deploy enterprise telemetry using OpenTelemetry standards with machine learning-based anomaly detection, custom performance visualization, and automated alerting to deliver predictive operational insights and support proactive maintenance actions.

Answer: A

Question: 96

CertyIQ

You are implementing Agentic AI within an Enterprise AI Factory. You are focused on the operation and scaling of the agentic systems including each of the Enterprise AI Factory components.

Which observability strategy involves providing detailed insights into the system's performance? (Choose two.)

- A. Detailed model and application tracing for identifying performance bottlenecks.
- B. Centralized logging to track system events.
- C. Continuous monitoring of key metrics using OpenTelemetry (OTEL).
- D. Artifact repository used by the AI agents where all the system performance metrics are stored.

Answer: AC

Question: 97

CertyIQ

Your support agent frequently fails to complete tasks when third-party tools return unexpected formats. Which solution improves resilience against these failures?

- A. Add robust schema validation and exception handling for all tool outputs
- B. Use deterministic temperature settings for all generations
- C. Reduce the number of tools available to avoid bad integrations
- D. Re-train the model to avoid the use of third-party tools entirely

Answer: A

Question: 98

CertyIQ

Which two error handling strategies are MOST important for maintaining agent reliability in production environments? (Choose two.)

- A. Circuit breaker patterns for external service calls
- B. Immediate failure propagation to users with verbose logging
- C. Automatic retry with exponential backoff for transient failures
- D. Immediate system shutdown for error handling

Answer: AC

Question: 99

CertyIQ

A customer service agent sometimes fails to complete multi-step workflows when APIs respond slowly or inconsistently.

Which approach most effectively increases robustness when working with unreliable APIs?

- A. Restrict available tools to reduce decision complexity
- B. Add retries with exponential backoff and set request timeouts
- C. Cache recent API results to limit unnecessary repeated calls
- D. Adjust generation parameters to produce more predictable responses

Answer: B

Question: 100

CertyIQ

You are building an agent that performs financial analysis by retrieving and processing structured data from a client's internal SQL database. The agent must handle occasional connection errors and retry the query up to a few times before failing gracefully.

Which approach best meets these requirements?

- A. Use structured tool calls with built-in retry handling and timed delays inside the tool wrapper
- B. Use few-shot prompting to guide the agent's conversation flow and manually retry failed API responses
- C. Use a reactive agent pattern that retries the query after a user confirms a retry attempt
- D. Use memory to track the number of failed attempts and apply it in later retries

Answer: A

Question: 101

CertyIQ

A large enterprise is preparing to roll out its AI-powered customer support agents worldwide. To maintain high availability and reliability, the operations team must select the best approach for monitoring, updating, and managing all agent instances across different locations.

Which solution most effectively ensures reliable operation and simplified management of large-scale agent deployments?

- A. Establishing centralized monitoring and automated deployment pipelines to oversee agent health, trigger updates, and manage rollbacks across all environments
- B. Allocating a dedicated support team to monitor agent logs and perform manual restarts to ensure human interaction in the data flywheel
- C. Scheduling updates and health checks on an annual basis to minimize service disruptions and ensure agent health, trigger updates, and manage rollbacks across all environments
- D. Provide separate monitoring tools and manual updates at each regional deployment for greater local control of agent health, trigger updates, and manage rollbacks across all environments

Answer: A

Question: 102

CertyIQ

You've deployed an agent that helps users troubleshoot technical issues with their devices. After several weeks in production, user feedback indicates a decline in response accuracy, especially for newer issues.

Which monitoring method is most appropriate for identifying the root cause of declining agent performance?

- A. Review output token counts across sessions to detect unusual model behavior
- B. Analyze logs of tool usage frequency and error rates during inference
- C. Compare average prompt length over time to analyze common input patterns
- D. Schedule a weekly re-deployment cycle to reset the model and improve freshness

Answer: B

Question: 103

CertyIQ

After deploying a financial assistant agent, users report occasional inconsistencies in how transactions are categorized.

What is the best first step for diagnosing the issue?

- A. Review and modify prompt temperature to enhance precision
- B. Review and retrain the model with more financial datasets
- C. Implement agent memory reset after each session
- D. Review tool call inputs and outputs in recent session logs

Answer: D

Question: 104

CertyIQ

Which two validation approaches are MOST critical for ensuring agent reliability in production deployments? (Choose two.)

- A. User satisfaction surveys as the primary quality metric
- B. Performance testing during development phases
- C. Structured output validation with Pydantic schemas
- D. Random sampling of agent interactions for manual review
- E. Automated consistency checking across multiple agent runs

Answer: CE

Question: 105

CertyIQ

You are implementing a RAG (Retrieval-Augmented Generation) solution.

What is the primary purpose of implementing semantic guardrails within a RAG system?

- A. To establish rules and constraints based on the meaning of user queries and generated responses.
- B. To eliminate all potential harmful entries from the vector database.
- C. To automatically translate all LLM responses into multiple languages for improved user comprehension.

D. To filter out all queries containing specific keywords that have been flagged as problematic.

Answer: A

Question: 106

CertyIQ

When implementing security measures for enterprise agentic systems using NVIDIA'S NeMo Guardrails, which approach provides the most comprehensive protection?

- A. Input sanitization at the user interface level
- B. Multi-layered guardrails with content moderation, output filtering, and behavioral monitoring
- C. Rule-based content filtering with predefined patterns
- D. User authentication and authorization controls

Answer: B

Question: 107

CertyIQ

Your team has built an agent using LangChain and needs to implement guardrails for deployment in a production environment.

Which approach represents the MOST effective integration of NVIDIA NeMo Guardrails?

- A. Rebuild the agent using only NeMo Guardrails, thereby reconstructing the LangChain implementation with enhanced safety controls and production-ready guardrail integration.
- B. Wrap the LangChain agent with NeMo Guardrails configuration while maintaining the existing workflow architecture and preserving current development investments.
- C. Configure input filtering to address safety requirements, integrating guardrail mechanisms focused on data validation and moderation within the current framework.
- D. Run the LangChain agent in parallel with NeMo Guardrails, allowing comparison of outputs between systems for comprehensive safety validation and performance optimization.

Answer: B

Question: 108

CertyIQ

This question addresses important concerns in the field of AI ethics and compliance, particularly as organizations develop more autonomous AI agents. Implementing effective guardrails against bias, ensuring data privacy, and adhering to regulations are essential components of responsible AI development.

Which of the following statements accurately describes how RAGAS (Retrieval Augmented Generation Assessment) can be utilized for implementing safety checks and guardrails in agentic AI applications?

- A. RAGAS cannot evaluate all safety aspects independently but provides metrics like Topic Adherence and Agent Goal Accuracy that serve as guardrails.
- B. RAGAS can only evaluate the quality of document retrieval but has no applications for safety guardrails in agentic systems.
- C. RAGAS is exclusively designed for hallucination detection and cannot evaluate other safety aspects of

agentic applications.

D. RAGAS can only be used in conjunction with other guardrail frameworks like NeMo and cannot function independently.

Answer: A

Question: 109

CertyIQ

Your team has deployed a generative agent for internal HR use, including summarizing candidate resumes and suggesting interview questions. After deployment, you've noticed that the model occasionally associates certain names or genders with particular roles.

Which mitigation strategy is the most effective and scalable for reducing this type of bias in agent outputs?

- A. Adjust system prompts to explicitly instruct the agent to avoid assumptions based on demographic features
- B. Randomly replace names in prompts to reduce identity correlation
- C. Add more training examples to the training dataset and re-train the model
- D. Implement guardrails to prevent outputs referencing protected attributes

Answer: D

Question: 110

CertyIQ

When analyzing safety violations in a financial advisory agent that uses NeMo Guardrails, which evaluation approach best identifies gaps in guardrail coverage?

- A. Apply keyword- and rule-based validation methods to confirm compliance with policy terms and common risk conditions.
- B. Analyze violation patterns, test adversarial prompts, measure guardrail activation, and align policies with observed failures.
- C. Conduct functional testing with representative user inputs to verify policy enforcement in typical usage scenarios.
- D. Monitor overall guardrail activations and system logs to assess operational behavior across different interaction types.

Answer: B

Question: 111

CertyIQ

You are developing a RAG solution and have decided to use a classifier branch as part of your semantic guardrail system to assess the risk of generated text.

Which of the following is a key benefit of using a classifier branch compared to solely relying on prompt filtering?

- A. Since a classifier branch does not require training, it can identify potentially problematic content.
- B. Classifier branches primarily focus on detecting factual inaccuracies, rather than stylistic or harmful language.
- C. Classifier branches can automatically adapt to new forms of harmful language.

D. Classifier branches eliminate the need for human oversight, thereby automating the safety process.

Answer: C

Question: 112

CertyIQ

You're deploying a healthcare-focused agentic AI system that helps doctors make treatment recommendations based on patient records. The agent's reasoning is not exposed to users, and its decisions sometimes differ from clinical guidelines.

What safety and compliance mechanisms should be in place? (Choose two.)

- A. Allow overrides by human doctors to maintain accountability
- B. Require model explainability or traceability for all outputs
- C. Prioritize autonomous speed of decision over explainability
- D. Exempt the model from compliance if it improves outcomes
- E. Obfuscate decision logic to protect proprietary methods

Answer: AB

Question: 113

CertyIQ

An AI Engineer is analyzing a production agentic AI system's compliance with responsible AI standards.

Which evaluation approaches effectively identify potential safety vulnerabilities and ethical risks in multi-agent workflows? (Choose two.)

- A. Emphasize latency metrics and throughput performance as key evaluation factors for safety vulnerabilities, providing a baseline for operational measures and resource allocation.
- B. Implement comprehensive audit trails using NVIDIA NeMo Guardrails with semantic similarity checks, tracking agent decisions across conversation flows and evaluating policy violations through automated compliance scoring.
- C. Use user feedback as a primary signal for risk identification, emphasizing post-deployment observations and qualitative experience reports alongside operational monitoring.
- D. Deploy multi-layered evaluation combining bias detection metrics (demographic parity, equalized odds) with adversarial testing to probe agent responses for harmful outputs across diverse user populations

Answer: BD

Question: 114

CertyIQ

You are deploying an AI-driven applicant-screening agent that analyzes candidate resumes and social-media data to recommend top applicants. Due to anti-discrimination laws and corporate policy, the system must mitigate bias against protected groups, maintain an audit trail of decisions, and comply with GDPR (including data minimization and explicit consent).

Which of the following strategies is most effective for ensuring your screening agent both mitigates bias in its recommendations and complies with data-privacy regulations?

- A. Perform a post-deployment GDPR and bias audit and process raw personal data as received.

- B. Pseudonymize protected attributes, implement fairness-aware debiasing, maintain an audit trail, and enforce GDPR data-minimization and consent.
- C. Encrypt all candidate data at rest and in transit, remove protected attributes from analysis, and conduct manual bias checks on recommendations.
- D. Exclude gender and ethnicity fields during training, use a generic privacy policy for consent, and do not maintain audit logs or apply targeted debiasing.

Answer: B

Question: 115

CertyIQ

A health assistant agent has been running on production environment for several weeks. The compliance team wants to audit how personal health data has been processed. Which operational feature supports this requirement?

- A. Adding more prompt examples to clarify privacy rules
- B. Masking all output with a profanity and PII detector
- C. Increasing model temperature for diverse interpretations
- D. Enabling full session logging with audit trail metadata

Answer: D

Question: 116

CertyIQ

You are designing an AI-powered drafting assistant for contract lawyers. The assistant suggests standard clauses and highlights potential risks based on past agreements. Senior attorneys must review, accept, modify, or reject each suggestion, see why a clause was recommended, and provide feedback to help improve the assistant. Which design feature is most critical for enabling effective human-in-the-loop oversight, transparency, and trust?

- A. Display suggested clauses with links to additional details about provenance and risk highlighting in a side panel, allowing users to access more context as needed.
- B. Insert suggested clauses into the draft and highlight changes for review at the end, inviting users to provide detailed feedback on clauses they wish to flag for improvement.
- C. Present batch “accept all” or “reject all” controls for suggested clauses, with explanations and feedback collected in a summary report after draft review.
- D. Show inline “why” explanations for each suggestion, highlight precedent and risk factors, and include accept/modify/reject controls with immediate feedback capture for model refinement.

Answer: D

Question: 117

CertyIQ

A development team is creating an AI assistant that interacts with employees to help manage schedules and tasks. The team wants to ensure users can easily provide feedback, understand the agent’s decisions, and intervene when necessary to maintain control and trust. Which practice best supports effective human oversight and interaction with the AI agent?

- A. Continuously collecting and integrating user feedback throughout the agent's lifecycle to drive ongoing improvements
- B. Incorporating user review stages before finalizing agent decisions to maintain accountability
- C. Enabling flexible user interactions beyond predefined commands to accommodate diverse needs
- D. Designing intuitive user interfaces with integrated feedback loops and transparent explanations of agent decisions

Answer: D

Question: 118

CertyIQ

A logistics company is implementing an agentic AI system for supply chain optimization that manages inventory levels, predicts demand, and automatically reorders supplies across multiple warehouses. Supply chain managers need to monitor AI decisions, understand the reasoning behind inventory recommendations, and intervene when business conditions change rapidly. The system must present complex data analytics in an intuitive way that enables quick decision-making while providing detailed insights when needed. Managers have varying levels of technical expertise and need interfaces that support both high-level oversight and detailed analysis. Which user interface design approach would BEST support effective human oversight of this complex multi-agent supply chain system?

- A. Develop a comprehensive dashboard with AI decision summaries, drill-down access to underlying data sets, and segmented performance metrics to enable targeted analysis of supply chain operations.
- B. Create separate specialized interfaces tailored to specific user roles, allowing managers to view AI-driven recommendations with drill-down options for role-specific details, but without a unified interface for cross-role collaboration.
- C. Create a layered interface featuring intuitive summaries, drill-down capabilities for detailed analysis, contextual explanations of AI decisions, and clear intervention controls with impact visualization and decision support tools.
- D. Create a streamlined interface presenting only high-level AI decisions and simplified recommendations, with drill-down views limited to basic historical trends for quick reference.

Answer: C

Question: 119

CertyIQ

An AI Engineer has deployed a multi-agent system to manage supply chain logistics. Stakeholders request greater insight into how the agents decide on actions across tasks. Which approach would best improve decision transparency without modifying the underlying model architecture?

- A. Gather structured user evaluations after each completed subtask
- B. Generate visual summaries of attention patterns for every decision
- C. Record a step-by-step reasoning log throughout each agent workflow
- D. Retain and share the full sequence of task instructions with stakeholders

Answer: C

Question: 120**CertyIQ**

When analyzing a customer service agentic system's performance degradation over time, which evaluation approach most effectively identifies opportunities for human-in-the-loop intervention to improve agent decision-making transparency and user trust?

- A. Monitor only final task completion rates without examining intermediate decision points, user interaction patterns, or opportunities for beneficial human intervention during agent conversations
- B. Implement multi-stage evaluation tracking decision confidence scores, user correction patterns, intervention effectiveness, and explainability-satisfaction correlations
- C. Rely on periodic manual reviews of random conversation samples without systematic tracking of intervention effectiveness, decision transparency, or user trust indicators
- D. Collect anonymous usage statistics without capturing specific decision rationales, user feedback on agent explanations, or transparency improvement opportunities for trust building

Answer: B**Question: 121****CertyIQ**

A medical diagnostics company is deploying an agentic AI system to assist radiologists in analyzing medical imaging. The system must provide AI-generated preliminary diagnoses and allow radiologists to review, modify, and approve all recommendations before patient treatment decisions. Human expertise should remain central, with detailed records of human interventions and decision rationales maintained.

Which approach would best balance human oversight with AI support in a safety-critical setting?

- A. Design an interactive system that presents AI analysis with confidence scores, allows radiologists to review evidence, modify recommendations, and requires explicit approval with documented reasoning for all decisions.
- B. Design a fully automated system that presents final diagnoses to radiologists for simple approval or rejection, minimizing human interaction to improve efficiency and reduce decision fatigue.
- C. Design a passive monitoring system where AI makes decisions while humans observe without ability to intervene, focusing on post-decision evaluation and quality assurance.
- D. Design a simple notification system that alerts radiologists only when AI confidence falls below predetermined thresholds, otherwise allowing autonomous operation without human review or documentation.

Answer: A**Question: 122****CertyIQ**

A development team is building an AI agent capable of autonomously planning and executing multi-step tasks while retaining context and learning from past interactions.

Which practice is most important to enable the agent to effectively manage long-term memory and complex tasks?

- A. Use basic rule-based decision methods that emphasize fast responses over adaptive planning.
- B. Apply short-term memory approaches that handle each interaction independently of previous ones.
- C. Implement memory mechanisms for context retention and apply chain-of-thought prompts to enhance reasoning.
- D. Reduce planning features and memory management to keep the system streamlined.

Answer: C

Question: 123

CertyIQ

You are implementing a RAG (Retrieval-Augmented Generation) solution. What is the primary purpose of implementing semantic guardrails within a RAG system?

- A. To filter out all queries containing specific keywords that have been flagged as problematic.
- B. To automatically translate all LLM responses into multiple languages for improved user comprehension.
- C. To eliminate all potential harmful entries from the vector database.
- D. To establish rules and constraints based on the meaning of user queries and generated responses.

Answer: D

Question: 124

CertyIQ

What is RAG Fusion primarily designed to achieve?

- A. Minimizing the need for retrieval, allowing the LLM to generate responses directly from its internal knowledge.
- B. Creating a separate, dedicated database for storing all the retrieved chunks.
- C. Automatically translating and integrating all retrieved chunks into a single language.
- D. Blending information from multiple retrieved chunks into a single response generated by the LLM.

Answer: D

Question: 125

CertyIQ

A customer service agentic AI is designed to resolve billing inquiries. It consistently resolves inquiries accurately and efficiently. However, a significant number of customers are reporting frustration due to the agent's tendency to repeatedly ask for the same information (account number, address) during each interaction, even after it's already been provided.

Which evaluation method would be most effective for addressing this issue?

- A. Adjusting the agent's reward function to prioritize speed of resolution over customer satisfaction.
- B. Implementing a "conversational flow" analysis to optimize the order of questions asked during each interaction.
- C. Analyzing the agent's dialogue transcripts to identify patterns in its questioning techniques.
- D. Increasing the agent's processing speed to reduce the time it takes to handle each inquiry and increase customer satisfaction.

Answer: C

Question: 126

CertyIQ

A company plans to launch a multi-agent system that must serve thousands of users simultaneously. The team

needs to ensure the system remains reliable, scales efficiently as demand increases, and operates in a cost-effective manner.

Which approach is most effective for achieving robust and scalable deployment of an agentic AI system in production?

- A. Deploying all agents on a single server with ongoing performance monitoring to maximize hardware utilization
- B. Orchestrating agents using containerization platforms, combined with load balancing and ongoing performance monitoring
- C. Running agents without load balancing to reduce infrastructure complexity and achieve robust and scalable deployment of an agentic system
- D. Establishing a continuous monitoring framework to track system performance and adapt resources as usage patterns evolve

Answer: B

Question: 127

CertyIQ

An agent is tasked with solving a series of complex mathematical problems that require external tools to find information. It often struggles to keep track of intermediate steps and reasoning.

Which prompting technique would be MOST effective in improving the agent's clarity and reducing errors in its reasoning?

- A. ReAct
- B. Symbolic Planning
- C. Zero-shot CoT
- D. Multi-Plan Generation

Answer: A

Question: 128

CertyIQ

A logistics company is implementing an agentic AI system for supply chain optimization that manages inventory levels, predicts demand, and automatically reorders supplies across multiple warehouses. Supply chain managers need to monitor AI decisions, understand the reasoning behind inventory recommendations, and intervene when business conditions change rapidly. The system must present complex data analytics in an intuitive way that enables quick decision-making while providing detailed insights when needed. Managers have varying levels of technical expertise and need interfaces that support both high-level oversight and detailed analysis.

Which user interface design approach would BEST support effective human oversight of this complex multi-agent supply chain system?

- A. Develop a comprehensive dashboard with AI decision summaries, drill-down access to underlying data sets, and segmented performance metrics to enable targeted analysis of supply chain operations.
- B. Create a layered interface featuring intuitive summaries, drill-down capabilities for detailed analysis, contextual explanations of AI decisions, and clear intervention controls with impact visualization and decision support tools.
- C. Create a streamlined interface presenting only high-level AI decisions and simplified recommendations, with drill-down views limited to basic historical trends for quick reference.

Answer: B

Question: 129**CertyIQ**

A company is deploying an AI-powered customer support agent that integrates external APIs and handles a wide range of customer inputs dynamically.

Which of the following strategies are appropriate when designing an AI agent for dynamic conversation management and external system interaction?

Pick the 2 correct responses below

- A. Using rule-based logic as the primary framework to maintain consistency in agent decisions.
- B. Implementing retry logic for API failures to ensure robustness in external communications.
- C. Integrating a feedback loop from user interactions to iteratively improve agent behavior.
- D. Preferring hardcoded responses for frequent queries to deliver reliable and low-latency answers.
- E.

Answer: B,C

Question: 130**CertyIQ**

In designing an AI workflow which of the following best describes a comprehensive approach to improving the performance of AI agents?

- A. Monitoring agents' throughput and time-to-first-token from the scoring engine
- B. Implementing benchmarking pipelines, deploying physical agents and monitoring user engagement metrics
- C. Implementing benchmarking pipelines and incorporating a dynamic dataset for a real-time fall-back
- D. Implementing benchmarking pipelines, collecting user feedback, and tuning model parameters iteratively

Answer: D

Question: 131**CertyIQ**

You are implementing a RAG (Retrieval-Augmented Generation) solution.

What is the primary purpose of implementing semantic guardrails within a RAG system?

- A. To filter out all queries containing specific keywords that have been flagged as problematic.
- B. To automatically translate all LLM responses into multiple languages for improved user comprehension.
- C. To eliminate all potential harmful entries from the vector database.
- D. To establish rules and constraints based on the meaning of user queries and generated responses.

Answer: D

Question: 132**CertyIQ**

When evaluating a customer service agent's resilience to API failures and network issues, which analysis methods effectively identify weaknesses in error handling and retry mechanisms?

Pick the 2 correct responses below

- A. Conduct failure injection testing with varied error types (timeouts, rate limits, malformed responses) while monitoring recovery patterns and fallback behavior.

- B. Implement retry mechanisms that standardize recovery attempts across scenarios, emphasizing consistency in handling errors.
- C. Analyze retry logic for exponential backoff patterns, retry limits, and circuit breaker integration to prevent cascading failures in distributed systems.
- D. Use fixed retry intervals to avoid the pitfalls of dynamic tuning, keeping retry timing consistent across different error conditions.
- E. Test under normal network conditions to establish baseline behavior, comparing results against production performance during degraded service scenarios.

Answer: A,C

Question: 133

CertyIQ

When evaluating an agent's integration with external tools and APIs for data retrieval and action execution, which analysis approaches effectively identify reliability and performance issues?
Pick the 2 correct responses below

- A. Connect to external APIs with standard procedures and monitor request and response exchanges to isolate the analysis of integration reliability and effectiveness.
- B. Design integration tests simulating API version changes, schema modifications, and backward compatibility scenarios to ensure reliable tool connections across updates.
- C. Use static API endpoints and parameters configured during development, allowing consistent and effective agent integration across predictable workflows.
- D. Implement comprehensive API call tracing with latency measurement, success rates per endpoint, and correlation analysis between tool failures and task completion.

Answer: A,B

Question: 134

CertyIQ

A company is building an AI agent that must retrieve information from large document collections and client databases in real time. The team wants to ensure fast, accurate retrieval and maintain high data quality. Which approach best supports efficient knowledge integration and effective data handling for such an agent?

- A. Integrating client data sources as they already incorporate data quality checks or augmentation to speed up deployment
- B. Using traditional relational databases because they don't need specialized retrieval mechanisms for all data queries
- C. Relying on pre-trained models instead of connecting to external knowledge sources during inference
- D. Implementing retrieval-augmented generation (RAG) pipelines combined with vector databases to accelerate access to relevant information

Answer: D

Question: 135

CertyIQ

Your team notices a spike in failed tool calls from a deployed workflow agent after a recent API schema update. The agent still returns outputs, but many are irrelevant or incomplete. Which maintenance task should be prioritized to restore accurate behavior?

- A. Reset the agent's long-term memory and reinitialize logs.
- B. Update the tool function specifications and re-test action sequences.
- C. Increase model temperature to encourage tool exploration.
- D. Reduce tool retrieval vector similarity threshold to broaden context.

Answer: B

Question: 136

CertyIQ

You're evaluating the performance of a tool-using agent (e.g., one that issues API calls or executes functions). From the list below, what are two important features to evaluate?
Pick the 2 correct responses below

- A. Tool use accuracy
- B. Tool use rate
- C. Task completion rate
- D. Tokens per second

Answer: A,C

Question: 137

CertyIQ

You are building a customer-support chatbot that fetches user account data from an external billing API. During testing, the API sometimes returns timeouts or 500 errors. You want the agent to be resilient — retrying when appropriate but failing gracefully if the service is down.

Which strategy best handles intermittent failures in API calls while still ensuring a good user experience?

- A. Return a standard fallback message on failures to maintain conversation flow and reduce the risk of service interruptions for the user.
- B. Schedule retries using a fixed delay for all failure types, maintaining predictable timing and user notifications after each attempt.
- C. Implement exponential-backoff retries with a circuit breaker, and return a clear message to the user if all retries fail.
- D. Retry requests with a consistent short delay after each failure and notify the user as each retry takes place.

Answer: C

Question: 138

CertyIQ

An AI Engineer at a retail company is developing a customer support AI agent that needs to handle multi-turn conversations while keeping track of customers' previous queries, preferences, and unresolved issues across multiple sessions.

Which approach is most effective for managing context retention and enabling the agent to respond coherently in real time?

- A. Use a sliding window of recent conversation tokens in memory to track only the last few exchanges.
- B. Retrain the model periodically using historical logs to improve long-term contextual understanding.
- C. Implement a hybrid memory system with vector-based search and key-value storage to retrieve relevant past

interactions.

D. Increase the maximum context window size so the full conversation history is processed each time.

Answer: C

Question: 139

CertyIQ

When implementing tool orchestration for an agent that needs to dynamically select from multiple tools (calculator, web search, API calls), which selection strategy provides the most reliable results?

- A. LLM-based tool selection with structured tool descriptions and usage examples
- B. Random dynamic tool selection with retry mechanisms and usage examples
- C. Rule-based selection with predefined tool mappings and usage examples
- D. Configuration-based tool selection with manual specifications and usage examples

Answer: A

Question: 140

CertyIQ

A healthcare AI company is deploying diagnostic agents that process medical imaging and patient data. The system must deliver consistent sub-100ms inference times for critical diagnoses while supporting deployment across multiple hospital sites with different NVIDIA GPU configurations (from RTX 6000 workstations to DGX systems). The agents need to maintain high accuracy while being portable across different hardware environments and capable of running efficiently on various GPU memory configurations.

Which optimization strategy would deliver the BEST performance improvements while maintaining deployment flexibility across diverse NVIDIA hardware configurations?

- A. Deploy agents using model optimizations with post-training quantization with Nvidia NIM deployment for portable performance across different GPU platforms and memory configurations.
- B. Deploy agents with NVIDIA CUDA-optimized Docker containers using a sequential inference architecture that processes each layer individually with GPU-to-CPU memory transfers between operations to avoid memory issues.
- C. Deploy agents using NVIDIA NIM containers with CPU-optimized inference to avoid GPU memory constraints and ensure consistent performance across different hospital infrastructure configurations.
- D. Deploy models using NVIDIA TensorRT optimization in their original FP32 precision format without any quantization or memory optimization, requiring 32GB+ GPU memory across all deployment sites.

Answer: A

Question: 141

CertyIQ

Your agent is designed to manage tasks through a service management API. The API responds with detailed event logs, but these logs contain both metadata and structured data.

To ensure the agent correctly interprets and processes the data from these logs, what's the most prudent approach?

- A. Employ a specialized parser that adheres to the API's documentation, to insure strict adherence to structured data.
- B. Employing a modular design that allows the agent to dynamically adjust its parsing logic.

- C. Employ a specialized parser that extracts all data fields, regardless of their type.
- D. Using a human-in-the-loop approach, manually inspecting and interpreting each log entry.

Answer: A

Question: 142

CertyIQ

When analyzing an agent's failure to complete multi-step financial analysis tasks, which evaluation approach best identifies prompt engineering improvements needed for reliable task decomposition and execution?

- A. Focus primarily on response speed optimization as a primary focus over reasoning quality, step completion accuracy, and prompt clarity for complex analytical requirements.
- B. Test only final output accuracy as this will automatically include intermediate reasoning steps, decomposition quality, and prompt structure effectiveness for complex workflows.
- C. Rely on generic prompt templates which are by default already optimized for general use, instead of tailoring them to financial terminology, calculation needs, or specialized multi-step analysis patterns.
- D. Implement systematic prompt testing with chain-of-thought reasoning templates, step-by-step decomposition analysis, and success rate tracking across tasks of varying complexity.

Answer: D

Question: 143

CertyIQ

When analyzing suboptimal agent response quality after deployment, which parameter tuning evaluation methods effectively identify the optimal configuration adjustments?

Pick the 2 correct responses below

- A. Implement A/B testing frameworks comparing temperature, top-k, and top-p variations while measuring task-specific quality metrics and user satisfaction scores.
- B. Use production traffic directly for parameter experiments, enabling real-world insights and faster identification of impactful settings.
- C. Design ablation studies systematically varying individual parameters while holding others constant to isolate each parameter's impact on agent behavior and performance.
- D. Apply identical parameter settings across all agent types and tasks, promoting consistency and simplifying comparison across different use cases.
- E. Randomly adjust all parameters simultaneously, allowing for broader exploration of the parameter space in a shorter time frame.

Answer: A,C

Question: 144

CertyIQ

An AI architect at a national healthcare provider is maintaining an agentic AI system. The system must monitor model and system performance in real time, raise alerts on failures or anomalies, manage version control and rollback of diagnostic models, and provide transparent insight into agent behavior during patient care workflows. Which operational approach best supports these requirements using the NVIDIA AI stack?

- A. Optimize all models with TensorRT and use periodic manual log reviews and NVIDIA shell scripts for detecting service anomalies and managing rollback.

- B. Containerize each agent in NIM with basic health checks running on cron jobs, and manage version rollback by swapping prebuilt container images.
- C. Deploy agent models on NVIDIA Triton Inference Server with Prometheus and Grafana for performance alerting, and manage model lifecycle via NGC and the Triton model repository.
- D. Expose agents as stateless NVIDIA API endpoints and monitor activity through application logs, with model versions tracked in a Git-based script repository.

Answer: C

Question: 145

CertyIQ

When analyzing performance bottlenecks in a multi-modal agent processing customer support tickets with text, images, and voice inputs, which evaluation approach most effectively identifies optimization opportunities?

- A. Optimize each modality independently using dedicated profiling of cross-modal interactions, shared resource constraints, and pipeline execution strategies.
- B. Profile end-to-end latency across modalities, measure model switching overhead, analyze batch processing opportunities, and evaluate Triton's dynamic batching for multi-modal workloads.
- C. Measure total response time as this analyzes aggregated performance trends across modalities, model loading times, and opportunities for parallel execution.
- D. Extend evaluation to accuracy and quality metrics, incorporating resource usage patterns, latency observations, and their impact on user experience.

Answer: B

Question: 146

CertyIQ

You are designing an AI-powered drafting assistant for contract lawyers. The assistant suggests standard clauses and highlights potential risks based on past agreements. Senior attorneys must review, accept, modify, or reject each suggestion, see why a clause was recommended, and provide feedback to help improve the assistant. Which design feature is most critical for enabling effective human-in-the-loop oversight, transparency, and trust?

- A. Insert suggested clauses into the draft and highlight changes for review at the end, inviting users to provide detailed feedback on clauses they wish to flag for improvement.
- B. Display suggested clauses with links to additional details about provenance and risk highlighting in a side panel, allowing users to access more context as needed.
- C. Show inline "why" explanations for each suggestion, highlight precedent and risk factors, and include accept/modify/reject controls with immediate feedback capture for model refinement.
- D. Present batch "accept all" or "reject all" controls for suggested clauses, with explanations and feedback collected in a summary report after draft review.

Answer: C

Question: 147

CertyIQ

You're evaluating the RAG pipeline by comparing its responses to synthetic questions. You've collected a large set of similarity scores.

What's the primary benefit of aggregating these scores into a single metric (e.g., average similarity)?

- A. Aggregation identifies the specific chunks within the RAG pipeline that are contributing to the highest similarity scores.
- B. Aggregation provides a more accurate representation of the RAG pipeline's performance.
- C. Aggregation reduces the complexity of the evaluation process and allows for a more overall assessment of the pipeline's effectiveness.
- D. Aggregation eliminates the need for qualitative analysis of the RAG pipeline's responses.

Answer: C

Question: 148

CertyIQ

An autonomous vehicle company operates a multi-agent AI system across its fleet to process real-time sensor data, make driving decisions, and communicate with cloud infrastructure. The company needs fleet-wide monitoring to track GPU utilization, inference times, and memory usage, correlate performance with driving conditions and system load, and predict safety issues before they occur.

Which monitoring and observability approach would BEST meet these fleet-scale, safety-critical requirements?

- A. Implement layered application monitoring with distributed tracing, synthetic transaction monitoring, and custom dashboards to capture complex dependencies, transaction flow, and service-level performance trends across the fleet.
- B. Deploy NVIDIA NIM microservices with Prometheus integration, NVIDIA Nsight Systems profiling, and Kubernetes-native monitoring to provide detailed metrics, profiling, and container orchestration observability across the entire stack.
- C. Deploy enterprise telemetry using OpenTelemetry standards with machine learning-based anomaly detection, custom performance visualization, and automated alerting to deliver predictive operational insights and support proactive maintenance actions.
- D. Implement comprehensive APM solutions with real-time baselines, automated root cause analysis, and fleet management integration to coordinate operational insights and performance management across thousands of vehicles.

Answer: C

Question: 149

CertyIQ

Optimize agentic workflow performance with the NVIDIA Agent Intelligence Toolkit.

Your organization is building a complex multi-agent system that needs to connect agents built on different frameworks while maintaining optimal performance.

Which key features of the NVIDIA Agent Intelligence Toolkit would be MOST beneficial for this implementation?

- A. The toolkit focuses primarily on agent development but lacks evaluation capabilities.
- B. The toolkit is designed exclusively for NVIDIA framework agents and cannot integrate with other frameworks.
- C. The toolkit provides framework-agnostic integration ensuring reusability of components.
- D. The toolkit is limited to simple agent-to-agent communication but cannot orchestrate complex multi-agent workflows.

Answer: C

Question: 150**CertyIQ**

When evaluating a multi-agent customer service system experiencing unpredictable scaling costs and performance bottlenecks during peak hours, which analysis approaches effectively identify optimization opportunities for both infrastructure efficiency and service reliability?
Pick the 2 correct responses below

- A. Deploy agents with configurable scaling workflows, allowing analysis of resource adjustment strategies and their effects on service stability during variable demand periods.
- B. Scale agent infrastructure based on aggregate performance trends, using system-wide monitoring tools to identify broader optimization patterns across resources.
- C. Implement comprehensive workload profiling using NVIDIA Nsight to analyze GPU utilization patterns, identify underutilized resources, and optimize batch sizing for dynamic scaling with Kubernetes HPA.
- D. Deploy distributed tracing with cost attribution per agent type, correlating resource consumption with business value metrics to identify optimization opportunities in agent deployment strategies.
- E. Maintain consistent resource allocation across all service hours, for a more precise view of baseline traffic impact on long-term infrastructure efficiency.

Answer: A,C**Question: 151****CertyIQ**

An AI Engineer at an automotive company is developing an inventory restocking assistant for parts that must plan reordering of parts over multiple days, factoring in stock levels, predicted demand, and supplier lead time. Which approach best equips the agent for sequential decision-making?

- A. Hybrid supervised/RL-trained model using NeMo-Aligner for policy alignment
- B. Reinforcement learning sequence model such as NVIDIA's NeMo-RL framework
- C. Rule-based reorder strategy with fixed thresholds implemented via NVIDIA Triton Inference Server
- D. Reinforcement learning sequence model using only a custom PyTorch Decision Transformer

Answer: B**Question: 152****CertyIQ**

Your support agent frequently fails to complete tasks when third-party tools return unexpected formats. Which solution improves resilience against these failures?

- A. Reduce the number of tools available to avoid bad integrations
- B. Use deterministic temperature settings for all generations
- C. Re-train the model to avoid the use of third-party tools entirely
- D. Add robust schema validation and exception handling for all tool outputs

Answer: D**Question: 153****CertyIQ**

Which two error handling strategies are MOST important for maintaining agent reliability in production

environments?

Pick the 2 correct responses below

- A. Automatic retry with exponential backoff for transient failures
- B. Immediate system shutdown for error handling
- C. Circuit breaker patterns for external service calls
- D. Immediate failure propagation to users with verbose logging

Answer: A,C

Question: 154

CertyIQ

You're managing an agentic AI responsible for customer support ticket triage. The agent has been consistently accurate in routing tickets to the appropriate departments. However, a team leader has noticed a significant increase in the number of tickets requiring "escalation" - cases where the agent initially misclassified a complex issue as a simple, routine one, leading to delays and frustrated customers.

What would be an appropriate first step in resolving this issue?

- A. Analyzing the agent's decision-making process, focusing on the specific criteria it uses to classify tickets, and identifying potential biases or blind spots.
- B. Adjusting the agent's reward function to prioritize speed of resolution over accuracy, as a first step in analysis of the problem.
- C. Conducting a "red-teaming" exercise, having human agents deliberately create complex and ambiguous scenarios to analyze the agent's robustness.
- D. Increasing the agent's autonomy, granting it more decision-making power during triage to improve its efficiency.

Answer: A

Question: 155

CertyIQ

Your team has built an agent using LangChain and needs to implement guardrails for deployment in a production environment.

Which approach represents the MOST effective integration of NVIDIA NeMo Guardrails?

- A. Wrap the LangChain agent with NeMo Guardrails configuration while maintaining the existing workflow architecture and preserving current development investments.
- B. Run the LangChain agent in parallel with NeMo Guardrails, allowing comparison of outputs between systems for comprehensive safety validation and performance optimization.
- C. Rebuild the agent using only NeMo Guardrails, thereby reconstructing the LangChain implementation with enhanced safety controls and production-ready guardrail integration.
- D. Configure input filtering to address safety requirements, integrating guardrail mechanisms focused on data validation and moderation within the current framework.

Answer: A

Question: 156

CertyIQ

When analyzing user feedback patterns to improve a technical documentation agent, which evaluation methods

effectively translate feedback into actionable optimization strategies?

Pick the 2 correct responses below

- A. Collect broad user feedback as-is, enabling rapid accumulation of suggestions and diverse perspectives for potential future analysis.
- B. Implement feedback categorization systems grouping issues by type (accuracy, clarity, completeness) with quantitative impact scoring and improvement prioritization matrices
- C. Incorporate user suggestions rapidly to maximize responsiveness and demonstrate continuous adaptation to evolving user needs.
- D. Design iterative feedback loops with version tracking, A/B testing of improvements, and regression monitoring to ensure changes enhance rather than degrade performance

Answer: B,D

Question: 157

CertyIQ

A recently deployed agent sometimes outputs empty responses under heavy system load. Which system-level signal is most useful for diagnosing this issue?

- A. GPU memory utilization and server-side inference logs
- B. Number of tool function arguments returned per query
- C. Prompt injection detection rate over time
- D. Retrieval similarity thresholds in vector search

Answer: A

Question: 158

CertyIQ

When designing tool integration for an agent that needs to perform mathematical calculations, web searches, and API calls, which architecture pattern provides the most scalable and maintainable approach?

- A. Microservice-based tool architecture with standardized interfaces
- B. Monolithic tool handler with conditional logic for different tool types
- C. External tool services with manual configuration for each agent instance
- D. Embedded tool functions within the main agent code

Answer: A

Question: 159

CertyIQ

When designing complex agentic workflows that include both sequential and parallel task execution, which orchestration pattern offers the greatest flexibility?

- A. Linear pipeline orchestration with a fixed task sequence
- B. Event-driven orchestration that triggers tasks reactively, in series or in parallel
- C. Graph-based workflow orchestration incorporating conditional branches

Answer: C

Question: 160

CertyIQ

An e-commerce platform is implementing an AI-powered customer support system that handles inquiries ranging from simple FAQ responses to complex product recommendations and technical troubleshooting. The system experiences unpredictable traffic patterns with sudden spikes during sales events and varying complexity requirements. Simple questions comprise the majority of requests but require minimal compute, while complex product recommendations need sophisticated reasoning. The company wants to optimize costs while maintaining service quality across all query types.

Which approach would provide the MOST cost-optimized scaling strategy for this variable-workload, mixed-complexity environment?

- A. Deploy specialized NVIDIA NIM microservices with an LLM router to dynamically route requests to appropriate models based on complexity, combined with auto-scaling infrastructure that scales different model types independently.
- B. Deploy specialized NVIDIA NIM microservices on CPU-optimized infrastructure with auto-scaling capabilities to minimize hardware costs, while accepting longer inference times for cost optimization benefits.
- C. Deploy specialized NVIDIA NIM microservices using a single large model configuration that handles all agent functions on high-capacity GPUs, with auto-scaling infrastructure that maintains constant resource allocation across all traffic patterns.
- D. Deploy multiple specialized NVIDIA NIM microservices with identical high-capacity models across all available GPUs, implementing auto-scaling infrastructure without request complexity differentiation or dynamic model selection capabilities.

Answer: A

Question: 161

CertyIQ

You are designing a virtual assistant that helps users check weather updates via external APIs. During testing, the agent frequently calls the incorrect tools, often hallucinating endpoints or returning incorrect formats. You suspect the prompt structure might be the root cause of these failures.

Which prompt design best supports consistent tool invocation in this agent?

- A. Include tool names in natural language but without parameter examples
- B. Rely on the agent's internal knowledge to infer tool usage
- C. Provide only a generic system instruction with no examples
- D. Use structured prompt templates with few-shot tool usage examples

Answer: D

Question: 162

CertyIQ

You are creating a virtual assistant agent that needs to handle an increasingly wide range of tasks over an extended period.

What is the primary benefit of combining external storage (like RAG) with fine-tuning (embodied memory) in this context?

- A. To ensure the agent doesn't make any errors
- B. To accelerate the agent's initial response time
- C. To enhance long-term reasoning capabilities and adaptability

D. To eliminate the need for external knowledge

Answer: C

Question: 163

CertyIQ

Your team has deployed a generative agent for internal HR use, including summarizing candidate resumes and suggesting interview questions. After deployment, you've noticed that the model occasionally associates certain names or genders with particular roles.

Which mitigation strategy is the most effective and scalable for reducing this type of bias in agent outputs?

- A. Randomly replace names in prompts to reduce identity correlation
- B. Implement guardrails to prevent outputs referencing protected attributes
- C. Add more training examples to the training dataset and re-train the model
- D. Adjust system prompts to explicitly instruct the agent to avoid assumptions based on demographic features

Answer: B

Question: 164

CertyIQ

When implementing security measures for enterprise agentic systems using NVIDIA's NeMo Guardrails, which approach provides the most comprehensive protection?

- A. User authentication and authorization controls
- B. Multi-layered guardrails with content moderation, output filtering, and behavioral monitoring
- C. Rule-based content filtering with predefined patterns
- D. Input sanitization at the user interface level

Answer: B

Question: 165

CertyIQ

A financial services agentic AI is being used to automate initial customer onboarding. The agent is completing the process efficiently and accurately, but reviews of its conversations reveal it often uses overly formal and complex language that confuses customers.

Which type of evaluation is best suited to address this issue?

- A. Controlled user testing sessions to collect user feedback on the clarity and tone of responses
- B. Compliance review of the agent's access to regulatory guidelines and policy documentation
- C. Continuous user feedback collection, specifically gathering subjective assessments of the agent's communication style
- D. Statistical analysis of the agent's decision-making patterns to detect overly formal and complex response choices

Answer: A

Question: 166**CertyIQ**

An AI engineer at an oil and gas company is designing a multi-agent AI system to support drilling operations. Different agents are responsible for subsurface modeling, risk analysis, and resource allocation. These agents must share operational context, reason through interdependent planning steps, and justify their collaborative decisions using structured, transparent logic. The architecture must support memory persistence, sequential decision-making and chain-of-thought prompting across agents. Which implementation best supports this design?

- A. Use LangChain to coordinate third-party agent APIs and store shared information in external memory, with logic encoded in static prompt chains.
- B. Fine-tune separate NeMo models for each agent role using LoRA, with pre-scripted action flows deployed via TensorRT for latency reduction.
- C. Orchestrate NeMo agents via Triton, use vector memory for shared context, ReAct planning, and NeMo Guardrails for reasoning.
- D. Use stateless LLM endpoints behind an API gateway and pass shared prompts across agents to simulate context and reasoning.

Answer: C**Question: 167****CertyIQ**

What NVIDIA framework can be used to train a better agent?

- A. NeMo-RL
- B. NeMo Guardrails
- C. TensorRT-LLM

Answer: A**Question: 168****CertyIQ**

In your RAG deployment, you've identified a performance bottleneck in the retrieval phase – specifically, the time it takes to access the vector database. Which of the following optimization strategies is most aligned with micro-service best practices, considering your RAG architecture?

- A. Introduce a dedicated service responsible solely for querying the vector database and returning relevant chunks.
- B. Implement a "cache-and-check" mechanism where the retrieval microservice immediately returns the first matching chunk, regardless of relevance.
- C. Increase the size of the LLM model itself, because it will automatically accelerate the overall response time.
- D. Optimize the LLM prompt to be shorter and more concise, significantly reducing the computational load.

Answer: A**Question: 169****CertyIQ**

This question addresses important concerns in the field of AI ethics and compliance, particularly as organizations

develop more autonomous AI agents. Implementing effective guardrails against bias, ensuring data privacy, and adhering to regulations are essential components of responsible AI development.

Which of the following statements accurately describes how RAGAS (Retrieval Augmented Generation Assessment) can be utilized for implementing safety checks and guardrails in agentic AI applications?

- A. RAGAS cannot evaluate all safety aspects independently but provides metrics like Topic Adherence and Agent Goal Accuracy that serve as guardrails.
- B. RAGAS can only evaluate the quality of document retrieval but has no applications for safety guardrails in agentic systems.
- C. RAGAS can only be used in conjunction with other guardrail frameworks like NeMo and cannot function independently.
- D. RAGAS is exclusively designed for hallucination detection and cannot evaluate other safety aspects of agentic applications.

Answer: A

Question: 170

CertyIQ

A development team is creating an AI assistant that interacts with employees to help manage schedules and tasks. The team wants to ensure users can easily provide feedback, understand the agent's decisions, and intervene when necessary to maintain control and trust.

Which practice best supports effective human oversight and interaction with the AI agent?

- A. Continuously collecting and integrating user feedback throughout the agent's lifecycle to drive ongoing improvements
- B. Enabling flexible user interactions beyond predefined commands to accommodate diverse needs
- C. Designing intuitive user interfaces with integrated feedback loops and transparent explanations of agent decisions
- D. Incorporating user review stages before finalizing agent decisions to maintain accountability

Answer: C

Question: 171

CertyIQ

In a global financial firm, an AI Architect is building a multi-agent compliance assistant using an agentic AI framework. The system must manage short-term memory for multi-turn interactions and long-term memory for persistent user and policy context. It should enable contextual recall and adaptation across sessions using NVIDIA's tool stack.

Which architectural approach best supports these requirements?

- A. Leverage NVIDIA Triton Inference Server with dynamic batching to cache session-level inputs between inference calls, and use an external Redis store for long-term memory.
- B. Leverage NVIDIA NeMo Framework with modular memory management, integrating conversational state tracking, knowledge graphs, and vector store retrieval, while using LoRA-tuned models to adapt responses over time.
- C. Leverage RAPIDS cuDF for memory tracking by streaming multi-turn conversation logs as GPU-resident data frames, assuming transactional history can be recalled and reasoned over using dataframe operations.
- D. Rely exclusively on TensorRT to encode all prior knowledge into compiled model weights, allowing inference-only execution with no external memory dependencies across sessions.

Answer: B

Question: 172**CertyIQ**

A technology startup is preparing to launch an AI agent platform to serve clients with unpredictable usage patterns. They face periods of high user activity and low demand, so their deployment approach must minimize wasted resources during slow times and automatically allocate more resources during busy periods — all while keeping operational costs reasonable.

Given these requirements, which deployment strategy most effectively ensures both cost-effectiveness and adaptability for scaling agentic AI systems?

- A. Implementing autoscaling policies in a container orchestration environment to automatically adjust resources according to workload changes
- B. Monitoring system logs for usage patterns and making infrastructure changes after monthly analysis
- C. Scheduling periodic manual reviews to increase or decrease infrastructure based on predicted user numbers
- D. Using fixed-size virtual machine clusters to guarantee consistent resource allocation at all times

Answer: A**Question: 173****CertyIQ**

You are developing an agent that needs to perform a complex set of tasks repeatedly.

Why is periodic fine-tuning an important aspect of long-term knowledge retention for this type of agent?

- A. It eliminates the need for external storage like RAG.
- B. It guarantees the agent will produce the same output for the same input.
- C. It prevents the agent from becoming overly specialized to a single task.
- D. It prevents the agent from forgetting past successes and failures.

Answer: D**Question: 174****CertyIQ**

A social media company wants to expand its agentic system to support global users, minimize downtime, and ensure smooth operation during usage spikes. The team is considering various deployment and scaling strategies to achieve these goals.

Which solution most effectively supports reliable and scalable deployment for an agentic AI system serving a global user base?

- A. Integrating MLOps practices for continuous deployment and rapid model updates in production environments
- B. Using hardware profiling to optimize agent workloads for efficient GPU utilization across all deployed instances
- C. Implementing containerization with Docker to simplify deployment and streamline updates
- D. Designing a distributed system architecture with multi-region deployment, automated failover, and dynamic resource allocation

Answer: D

Question: 175**CertyIQ**

When analyzing safety violations in a financial advisory agent that uses NeMo Guardrails, which evaluation approach best identifies gaps in guardrail coverage?

- A. Analyze violation patterns, test adversarial prompts, measure guardrail activation, and align policies with observed failures.
- B. Monitor overall guardrail activations and system logs to assess operational behavior across different interaction types.
- C. Conduct functional testing with representative user inputs to verify policy enforcement in typical usage scenarios.
- D. Apply keyword and rule-based validation methods to confirm compliance with policy terms and common risk conditions.

Answer: A**Question: 176****CertyIQ**

Your agent's primary task is to collect data from a device management platform API. This API frequently returns data in JSON format, but the underlying schemas can evolve without detailed documentation. What's the most effective strategy for the agent to maintain consistent access to the data, despite potential schema changes?

- A. Building a fixed data extraction method based on the API's historical response patterns.
- B. Building a JSON schema validator that constantly monitors and enforces the current API schema.
- C. Manually inspecting the API response's structure and updating the agent's code when a change is detected.
- D. Building a flexible data mapping layer that can adapt to changes in the API response structure.

Answer: D**Question: 177****CertyIQ**

You are improving your Retrieval-Augmented Generation (RAG) pipeline, and you want to exploit the properties of the embedding model to improve the solutions. Within a RAG system, why is reranking retrieved chunks a crucial step after initial retrieval?

- A. Reranking improves retrieval accuracy, regardless of the initial retrieval method.
- B. Reranking allows for prioritizing chunks based on relevance to the semantic intent of the query.
- C. Reranking primarily focuses on optimizing the visual presentation of the retrieved chunks.
- D. Reranking's purpose is to reduce the overall length of the response generated by the LLM.

Answer: B**Question: 178****CertyIQ**

An agentic AI is tasked with generating marketing copy for various campaigns. It's consistently producing high-quality text and generating significant engagement. However, qualitative feedback from brand managers indicates that the content lacks a distinct "brand voice" and feels generic. Which of the following metrics would be most valuable for evaluating the agent's adherence to the brand's

established voice?

- A. A metric evaluating the agent's textual similarity to a formalized brand style guide, analyzing factors such as tone, approved vocabulary, and prescribed sentence structures.
- B. A metric tracking the average word count and sentence length of the agent's copy, focusing on stylistic efficiency as a potential proxy for brand alignment.
- C. A metric quantifying how frequently the agent's output is shared, liked, or reposted on major social platforms, using this as an indicator of effective brand representation.
- D. A metric assessing the agent's ability to tailor its language and messaging for distinct audience segments based on demographic and psychographic data.

Answer: A

Question: 179

CertyIQ

You are rolling out a multimodal conversational agent on NVIDIA's stack: the model is containerized as a TensorRT-LLM engine, served via Triton Inference Server behind NIM microservices for routing and scaling, and protected by NeMo Guardrails for safety and compliance. During early testing, end-to-end latency exceeds your target budget, and you need to tune batching, model precision, and guardrail checks while maintaining both throughput and enforcement of safety policies.

Which configuration change is most effective for reducing latency under these constraints while still enforcing NeMo Guardrails policies?

- A. Keep FP32 precision, increase batch size aggressively, and perform Guardrails checks in a downstream microservice after inference.
- B. Deploy separate Triton servers for model inference and guardrail validation, routing requests sequentially and merging outputs at the application layer.
- C. Quantize the TensorRT-LLM engine to INT8, disable dynamic batching, and invoke Guardrails checks synchronously within the inference path.
- D. Quantize the TensorRT-LLM engine to FP16, tune Triton's dynamic batching, and integrate NeMo Guardrails alongside inference to run policy checks in parallel.

Answer: D

Question: 180

CertyIQ

An AI Engineer has deployed a multi-agent system to manage supply chain logistics. Stakeholders request greater insight into how the agents decide on actions across tasks.

Which approach would best improve decision transparency without modifying the underlying model architecture?

- A. Generate visual summaries of attention patterns for every decision
- B. Gather structured user evaluations after each completed subtask
- C. Record a step-by-step reasoning log throughout each agent workflow
- D. Retain and share the full sequence of task instructions with stakeholders

Answer: C

Question: 181

CertyIQ

A Lead AI Architect at a global financial institution is designing a multi-agent fraud detection system using an agentic AI framework. The system must operate in real time, with distinct agents working collaboratively to monitor and analyze transactional patterns across accounts, retain and share contextual information over time, and escalate suspicious behaviors to a human fraud analyst when needed. Which architectural approach enables intelligent specialization, shared memory, and inter-agent coordination in a dynamic and evolving threat environment?

- A. Design an agentic workflow where each agent acts independently on isolated data slices with no inter-agent communication to reduce latency and model complexity.
- B. Design a modular multi-agent system where individual agents collaborate asynchronously using shared memory and structured messaging.
- C. Design a multi-agent system where individual agents collaborate synchronously using shared memory and structured messaging.
- D. Design monolithic LLM-based agents that handle all fraud detection tasks within a single loop, without modular roles or multi-agent coordination.
- E. Design a centralized rule-based service that checks all transactions against static fraud indicators and sends alerts when thresholds are exceeded.

Answer: B

Question: 182

CertyIQ

You are evaluating your RAG pipeline. You notice that the LLM-as-a-Judge consistently assigns high similarity scores to responses that contain irrelevant information.

What should you investigate as the most likely potential cause with the least development effort?

- A. The quality of the synthetic questions used for evaluation.
- B. The size of the knowledge base used to power the RAG pipeline.
- C. The prompt used to instruct the LLM-as-a-Judge to assess the response.
- D. The temperature setting used by the LLM during response generation.

Answer: C

Thank you

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